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Structural and mechanistic insight into ribosomal ITS2 RNA processing by nuclease-kinase machinery

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Abstract

Precursor ribosomal RNA (pre-rRNA) processing is a key step in ribosome biosynthesis and involves numerous RNases. A HEPN nuclease Las1 and a polynucleotide kinase Grc3 assemble into a tetramerase responsible for rRNA maturation. Here, we report the structures of full-length *Saccharomyces cerevisiae* and *Cyberlindnera jadinii* Las1-Grc3 complexes, and *Cyberlindnera jadinii* Las1. The Las1-Grc3 structures show that the central coiled coil domain of Las1 facilitates pre-rRNA binding and cleavage, while the Grc3 C-terminal loop motif directly binds to the HEPN active center of Las1 and regulates pre-rRNA cleavage. Structural comparison between Las1 and Las1-Grc3 complex exhibits that Grc3 binding induces conformational rearrangements of catalytic residues associated with HEPN nuclease activation. Biochemical assays identify that Las1 processes pre-rRNA at the two specific sites (C2 and C2'), which greatly facilitates rRNA maturation. Our structures and specific pre-rRNA cleavage findings provide crucial insights into the mechanism and pathway of pre-rRNA processing in ribosome biosynthesis.

eLife assessment

This study represents a **valuable** mechanistic contribution towards understanding how ribosomal RNA is processed during ribosome biogenesis. The biochemical evidence supporting the major conclusions is **solid**, but the structural biology data is **incomplete** and would benefit from more rigorous analysis. After further strengthening, this work would be of interest to cell biologists and biochemists working on ribosome biogenesis.

Introduction

Ribosomes are large molecular machines assembled from numerous proteins and RNAs that are responsible for protein synthesis in cells (Anger et al., 2013; Gasse, Flemming, & Hurt, 2015; Khatter, Myasnikov, Natchiar, & Klaholz, 2015). In eukaryotes, ribosome biosynthesis is tightly coupled to cell growth and cell cycle progression and is critical for regulating normal

cell size and maintaining cell cycle progression (Castle et al., 2013). Ribosome biosynthesis is an extremely complicated process involving about 200 assembly and processing factors, which are involved in a series of continuous assembly and processing reactions such as ribosome protein folding, modification, assembly, and precursor rRNA (pre-rRNA) processing (Gasse et al., 2015; Lafontaine, 2015; Pillon, Sobhany, Borgia, Williams, & Stanley, 2017; Wu et al., 2016). Mature ribosomes in yeast *Saccharomyces cerevisiae* contain 79 proteins and four RNAs (25S, 18S, 5.8S, and 5S rRNA) (Doudna & Rath, 2002; Wilson & Doudna, 2012; Woolford & Baserga, 2013). 5S rRNA is transcribed by RNA polymerase III, whereas 25S, 18S and 5.8S rRNA are cotranscribed by RNA polymerase I as a single long precursor (35S pre-rRNA) (Tomecki, Sikorski, & Zakrzewska-Placzek, 2017). Except 25S, 18S and 5.8S rRNA sequences, the 35S pre-rRNA also includes 5'-external transcribed spacer sequence (ETS), 3'-external transcribed spacer sequence, and two internal transcribed spacer sequences (ITS1 and ITS2) (Fromm et al., 2017). ITS1 located between 5.8S and 18S rRNA and ITS2 located between 5.8S and 25S rRNA (Coleman, 2003; Cote, Greer, & Peculis, 2002). The mature rRNAs are generated by a large number of endonucleases and exonucleases that remove these transcribed spacers step by step through multiple efficient and correct processing reactions (Granneman, Petfalski, & Tollervey, 2011). The pre-rRNA processing factors synergistically produce mature rRNAs and lead to an accurate and efficient assembly of mature ribosome in the nucleolus, which are key to cell survival (Pillon & Stanley, 2018). Mutations in the genes that encode these pre-rRNA processing factors are often lethal (Tomecki et al., 2017). Although numerous evolutionarily conserved protein factors have been found to be involved in the processing and modification of ribosomal RNA, the detailed pathways of these factors and their specific processing mechanisms are not well understood.

Las1 and Grc3 are highly conserved proteins and have recently been identified as core enzymes involved in processing and removing ITS2 spacer, which is a key step in the synthesis of 60S ribosomal subunit (Gasse et al., 2015; Schillewaert, Wacheul, Lhomme, & Lafontaine, 2012). Las1 is characterized as a nucleolar protein essential for ribosome biogenesis, as well as cell proliferation and cell viability in *Saccharomyces cerevisiae* (Castle, Cassimere, & Denicourt, 2012; Castle, Cassimere, Lee, & Denicourt, 2010; Doseff & Arndt, 1995). It is important to determine the role of Las1 in rRNA metabolic pathways and regulatory networks associated with ribosome biogenesis and cell proliferation. Recent studies have demonstrated that Las1 is an endoribonuclease that contains a HEPN (higher eukaryote and prokaryote nucleotide binding) domain responsible for rRNA processing (Pillon et al., 2020; Pillon et al., 2017). The HEPN domain must be dimerized to form an active nuclease, such as the Cas13 effectors in CRISPR immune defense systems, whose catalytic site is formed by two HEPN domains involved in non-specific cleavage of single-stranded RNA (Knott et al., 2017; Zhang et al., 2018). Interestingly, Las1 specifically cleaves at the C2 site within ITS2 and generates the 7S pre-rRNA and 26S pre-rRNA (Fernandez-Pevida, Kressler, & de la Cruz, 2015). It is not clear why Las1 HEPN nuclease specifically targets and cleaves ITS2 only at C2 site, and whether this cleavage depends on a specific sequence or secondary structure. In addition, the cleavage activity of Las1 is primarily dependent on another enzyme Grc3. Grc3 is a polynucleotide kinase responsible not only for Las1 nuclease activation, but also for nonspecific phosphorylation of the 5'-OH of the 26S pre-rRNA produced by Las1 cleavage, providing a signal for further processing by Rat1-Rai1 exonuclease (Gordon, Pillon, & Stanley, 2019; Xiang et al., 2009). Although Cryo-EM reveals cross-linked *Chaetomium thermophilum* Las1 and Grc3 assemble into a super-dimer, due to flexibility, critical structural information is missing for the coiled coil (CC) domain of Las1 and the N-terminal and C-terminal regions of Grc3 (Pillon et al., 2019). It remains unknown how Las1 and Grc3 coordinate with each other in terms of substrate binding and nuclease activation.

In this study, we identified that *S. cerevisiae* (Sc) Las1 endoribonuclease initially cleaves ITS2 in a step-by-step fashion at two specific sites, which greatly promotes the maturation of 25S rRNA. Additionally, we solved the crystal structures of full-length ScLas1-Grc3 complex and *Cyberlindnera jadinii* (Cj) Las1-Grc3 complex, as well as the high resolution structure of CjLas1 HEPN domain. Our structural and biochemical findings uncovered a detailed mechanism of polynucleotide kinase-mediated activation of HEPN nuclease, providing a molecular basis for clearly understanding the process of pre-rRNA processing and maturation in ribosome biosynthesis.

Results

Las1 cleaves ITS2 at two specific sites

HEPN endoribonucleases such as Cas13, Ire1, and RNase L are metal-independent RNA specific enzymes that efficiently cleave substrate RNA at multiple sites (Abudayyeh et al., 2016; Huang et al., 2014; Korennykh et al., 2009; Lee et al., 2008; Wang et al., 2014). Las1 is also identified as a HEPN-containing RNase, but it has only been found to initially cut ITS2 RNA at a single specific position (C2). To investigate whether there are other potential sites for Las1 cleavage in ITS2, we performed *in vitro* RNA cleavage assays using a 5'-Cy5- and 3'-Cy3-labeled 33-nt ITS2 RNA substrate (Figure 1A). Las1 shows weak or no detectable activity to ITS2 in the absence of Grc3, but exhibits robust activity in the presence of Grc3 (Figure 1B, Figure 1–figure supplement 1), revealing Grc3-dependent Las1 nuclease activation. Interestingly, we observed a single 5'-Cy5-labeled cleavage product, and two prominent 3'-Cy3-labeled cleavage products including a final product and an intermediate product (Figure 1B), suggesting that cleavage of ITS2 substrate occurs at two specific sites. The 33-nt ITS2 substrate RNA harbors the C2 site, which is located between nucleotides A140 and G141 (Figure 1A). In addition to the C2 site, there is another specific position in ITS2 that is able to be processed by Las1. ScLas1 cleaves the 33-nt ITS2 at C2 site to theoretically generate a 10-nt 5'-terminal product and a 23-nt 3'-terminal product (Figure 1A). Our merger data shows that the final 5'-terminal and 3'-terminal product bands are at nearly the same horizontal position on the gel (Figure 1B), indicating that they are similar in size. Therefore, we hypothesize that the 23-nt 3'-terminal product is an intermediate that can be further processed by Las1 at a specific site to produce two small products. We then mapped the cleavage products using the 33-nt ITS2 RNA without 5'-Cy5- and 3'-Cy3-label. We observed four cleavage bands of approximately 23-nt (P2), 14-nt (P3), 10-nt (P1), and 9-nt (P4) in length (Figure 1C). The 23-nt product (P2) is obviously an intermediate cleavage product, which is further cleaved to generate 14-nt and 9-nt products. Based on these observations, we identify that another cleavage site, which we designated as C2', is located between nucleotides G154 and C155. Las1 is able to process ITS2 at C2 and C2' sites in a step-by-step manner, resulting in a 9-nt 3'-end product and a 10-nt 5'-end product.

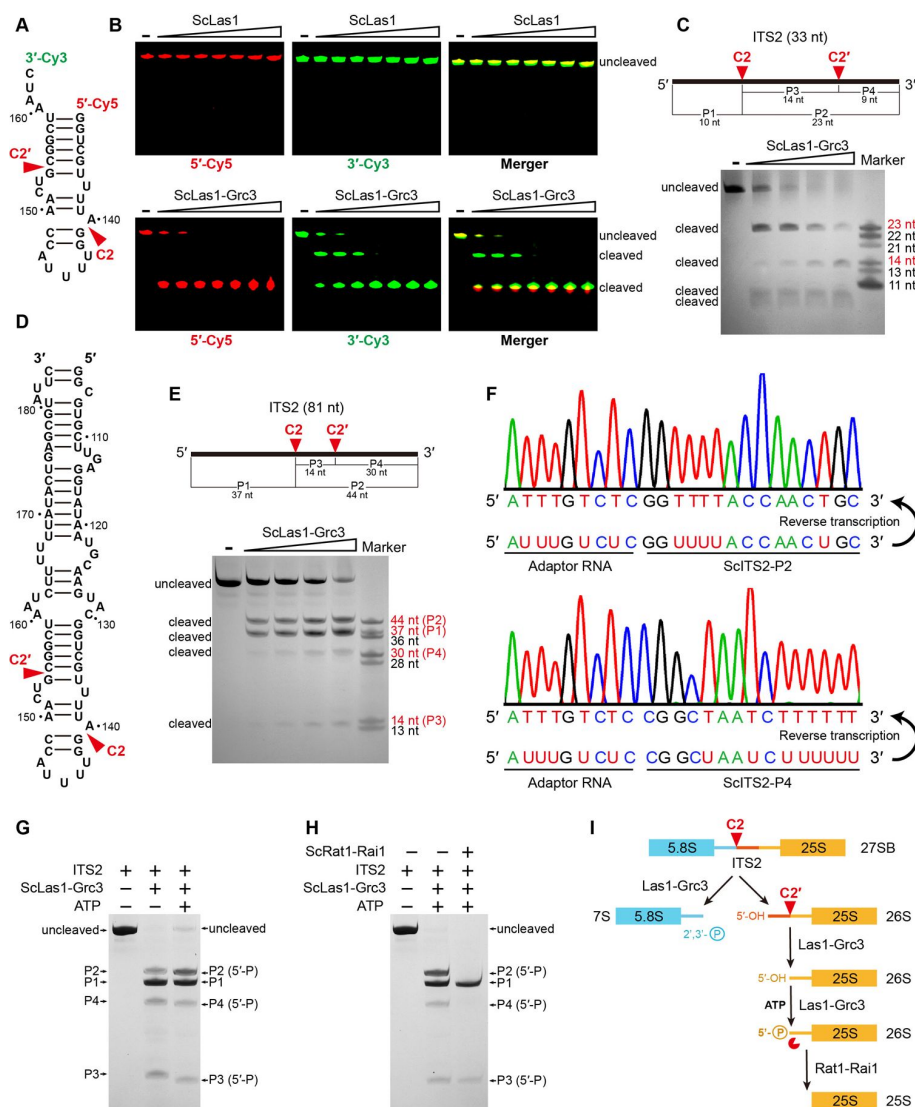


Figure 1.

ScLas1 specifically cleaves ITS2 at C2 and C2' sites.

(A) 33-nt ITS2 RNA with 5'-Cy5 and 3'-Cy3 labels. **(B)** *In vitro* RNA cleavage assay using 5'-Cy5 and 3'-Cy3 labeled 33-nt RNA. **(C)** *In vitro* RNA cleavage assay of unlabeled 33-nt RNA. **(D)** 81-nt ITS2 RNA. **(E)** *In vitro* RNA cleavage assay of unlabeled 81-nt RNA. **(F)** RNA-sequencing traces from ScLas1-cleaved ITS2 products P2 and P4. **(G)** RNA phosphorylation assay with ScLas1-Grc3 complex. **(H)** RNA degradation assay with ScRat1-Rai1 complex. **(I)** The ITS2 pre-rRNA processing pathway. The following figure supplements are available for Figure 1:

We obtained similar cleavage results with a longer 81-nt ITS2 RNA substrate (Figure 1D, E). To further confirm the cleavage site of C2', we then mapped the cleavage sites of the 81-nt ITS2 using reverse transcription coupling sequencing methods (Figure 1F). We used an adaptor RNA to separately link the product RNA fragments to form template RNAs. After reverse transcription and sequencing of these template RNAs, we obtained the accurate sequence information for products P2 and P4, revealing that the C2 site is located between nucleotides A140 and G141, and the C2' site is located between nucleotides G154 and C155.

The cleavage products P2, P3, and P4 are all phosphorylated by Grc3, as they all show a slight shift when in the presence of ATP in the reaction (Figure 1G). This indicates that Las1 generates 5'-OH terminus following both C2 and C2' cleavage. The phosphorylated products are further degraded by Rat1-Rai1 exonuclease, especially P1 and P2, which are completely degraded under given experimental conditions (Figure 1H).

In addition, Grc3 shows no cleavage activity to ITS2 (Figure 1-figure supplement 2), and mutations in the HEPN catalytic residues of Las1 abolish C2 and C2' cleavage (Figure 1-figure supplement 3), further confirming that both C2 and C2' cleavages are attributed to metal-independent Las1 RNase (Figure 1-figure supplement 4).

Overall structures of ScLas1-Grc3 complex and CjLas1-Grc3 complex

To elucidate how Las1 and Grc3 cooperate to direct ITS2 cleavage and phosphorylation, we determined the cryo-electron microscopy (cryo-EM) and crystal structures of full-length ScLas1-Grc3 complex at 3.07 Å and 3.69 Å resolution (Figure 2, Figure 2-figure supplement 1, Table 1 and Table 2), respectively. Cleavage assay indicates that ScGrc3 can activate ScLas1 nuclease well for ITS2 cleavage, while CjGrc3 exhibits relatively weak activation ability for CjLas1 nuclease (Figure 1B, Figure 1-figure supplement 1). To better understand the differences in the mechanisms of Grc3 activation of Las1 between different species, we also solved the cryo-EM and crystal structures of full-length CjLas1-Grc3 complex at 3.39 Å and 3.39 Å resolution (Figure 3, Figure 3-figure supplement 1, Table 1 and Table 2), respectively. To be noted, the crystal structures were solved by molecular replacement method with the cryo-EM maps of Las1-Grc3 complexes. Since the crystal structures have similar quality and resolve more structural information than the cryo-EM structures (Figure 3-figure supplement 2), the subsequent presentations and descriptions are based on the crystal structures.

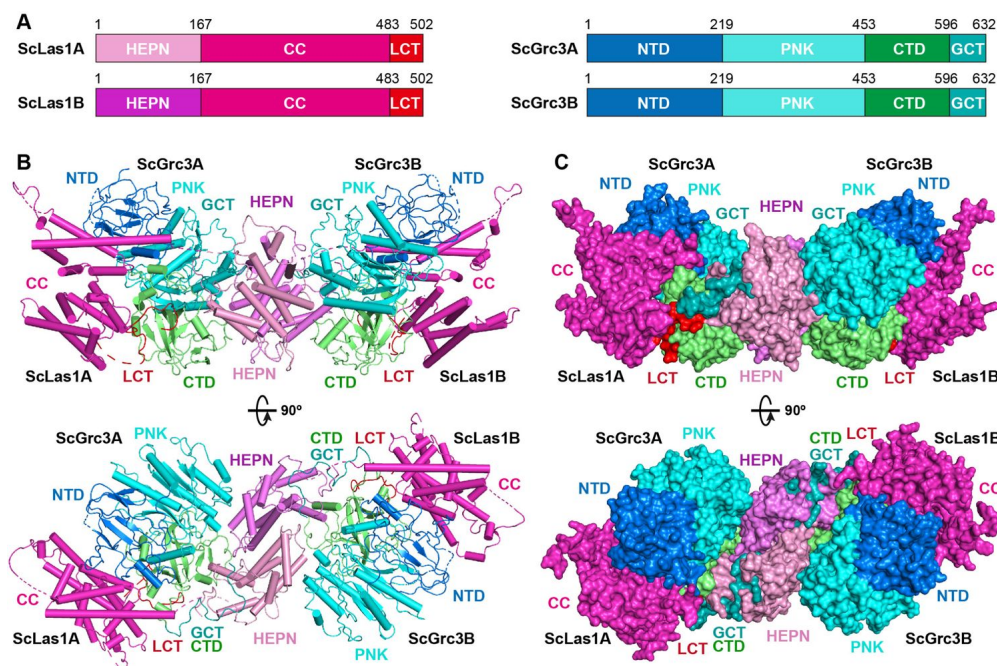


Figure 2.

Overall structure of ScLas1-Grc3 complex.

(A) Domain organization of ScLas1 and ScGrc3. (B) Ribbon representations of ScLas1-Grc3 complex. Color-coding used for Las1 and Grc3 is identical to that used in Fig. 2a. (C) Surface representations of ScLas1-Grc3 complex. Color-coding used for Las1 and Grc3 is identical to that used in Figure 2A. The following figure supplement is available for Figure 2:

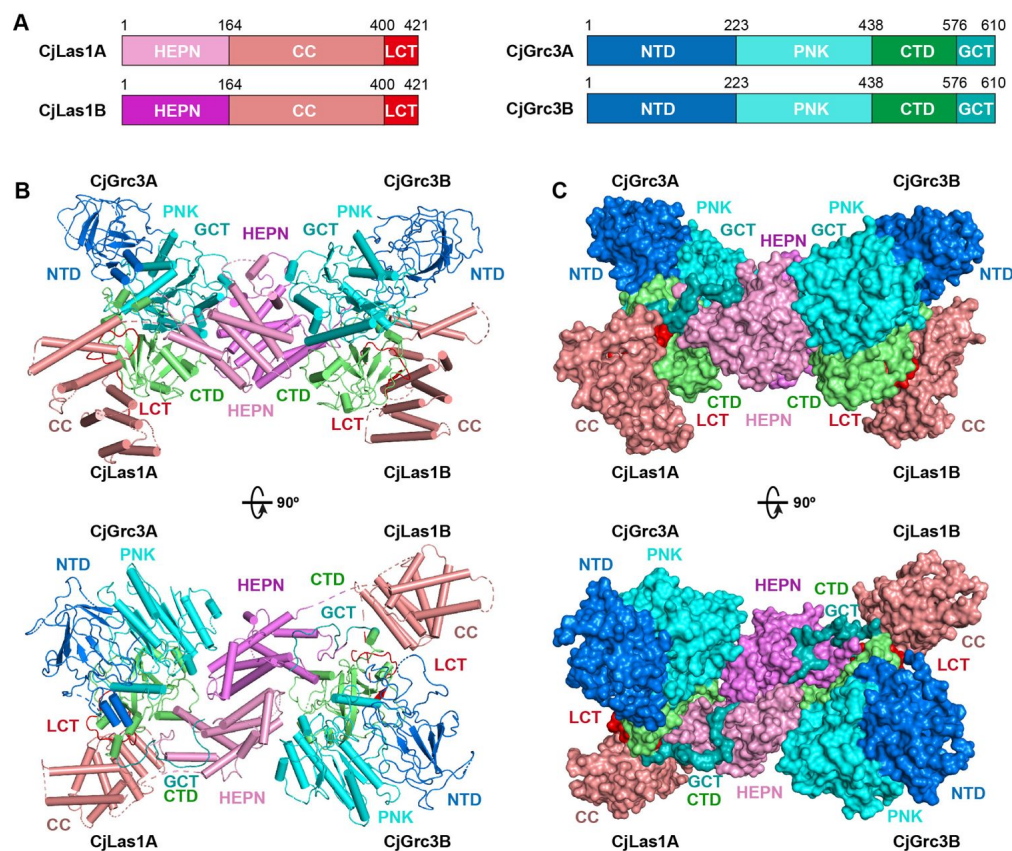


Figure 3.

Overall structure of CjLas1-Grc3 complex.

(A) Domain organization of CjLas1 and CjGrc3. (B) Ribbon representations of CjLas1-Grc3 complex. Color-coding used for Las1 and Grc3 is identical to that used in Fig. 3A. (C) Surface representations of CjLas1-Grc3 complex. Color-coding used for Las1 and Grc3 is identical to that used in Figure 3A. The following figure supplements are available for Figure 3:

Table 1.

Crystallographic data collection and refinement statistics.

	ScLas1-Grc3	CjLas1-Grc3	CjLas1
Data collection*			
Space group	C2	C222 ₁	P2 ₁ 2 ₁ 2 ₁
Cell dimensions			
<i>a</i> , <i>b</i> , <i>c</i> (Å)	233.6, 116.1, 159.3	152.6, 240.0, 237.0	51.5, 59.0, 158.7
α , β , γ (°)	90.0, 96.4, 90.0	90.0, 90.0, 90.0	90.0, 90.0, 90.0
Resolution (Å)	50.00-3.50 (3.56-3.50)	50.00-3.23 (3.29-3.23)	50.00-1.80 (1.83-1.80)
<i>R</i> _{merge}	0.298 (0.980)	0.344 (0.958)	0.103 (0.929)
<i>I</i> / σ <i>I</i>	4.8 (1.1)	4.3 (1.6)	22.0 (2.5)
Completeness (%)	98.8 (96.5)	99.9 (99.9)	97.7 (95.8)
Redundancy	4.5 (3.5)	7.5 (6.4)	10.1 (9.5)
Refinement			
Resolution (Å)	3.69	3.39	1.80
No. reflections	36,773	41,321	44,296
<i>R</i> _{work} / <i>R</i> _{free}	0.2798/0.3151	0.3041/0.3281	0.2120/0.2334
No. atoms			
protein	22,873	18,763	3,657
Water	180	386	212
<i>B</i> -factors (Å ²)			
Protein	95.5	125.6	23.3
Water	33.2	55.9	28.5
R.m.s. deviations			
Bond length (Å)	0.008	0.011	0.015
Bond angles (°)	1.516	1.785	1.500
Ramachandran plot			
Favored region	94.96	95.72	97.98
Allowed region	4.86	4.28	2.02
Outlier region	0.18	0.00	0.00

* Highest resolution shell is shown in parentheses.

Table 2.

Cryo-EM data collection and processing.

	ScLas1-Grc3	CjLas1-Grc3
Data collection and processing		
Microscope	Titan Krios	Titan Krios
Voltage (kV)	300	300
Camera	Gatan K3	Gatan K3
Magnification	105,000x	105,000x
Pixel size (Å)	0.82	0.82
Total exposure (e-/Å ²)	50	50
Exposure time (s)	3	3
Number of frames per exposure	30	30
Energy filter slit width (keV)	20	20
Data collection software	EPU	EPU
Defocus range (μm)	-1.3 - -2.7	-1.2 - -3
Number of micrographs	2,520	8,616
Number of initial particles	525,213	2,215,555
Symmetry	C2	C2
Number of final particles	264,341	523,843
Resolution (0.143 gold standard FSC, Å)	3.07	3.39
Local resolution range (Å)	2.8-4.8	2.8-4.8

Both structures reveal that Las1 and Grc3 assemble into a tetramer with two copies of each (Figure 2B, C and Figure 3B, C). The architecture of Las1-Grc3 tetramer complex resembles a butterfly with outstretched wings. Las1 consists of a relatively conserved N-terminal HEPN domain, a poorly conserved central coiled coil (CC) domain, and a short C-terminal tail motif (LCT) (Figure 2A, Figure 3A, Figure 3-figure supplement 3). Grc3 is composed of an N-terminal domain (NTD), a central polynucleotide kinase (PNK) domain, a C-terminal domain (CTD), and a short C-terminal loop motif (GCT). Two HEPN domains from two Las1 copies are tightly symmetrically stacked together to form a HEPN dimer that constructs the body of the butterfly (Figure 2B and Figure 3B). Two CC domains of Las1 and two Grc3 copies assemble into the left and right wings. The NTD and PNK domains of Grc3 constitute the forewing, while the CTD domain of Grc3 and the CC domain of Las1 form the hindwing. All domains are perfectly stacked together to build a compact and stable tetramer architecture (Figure 2C

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Las1 CC domain contributes to ITS2 binding and enhances cutting

In order to explore whether there are structural differences between ScLas1-Grc3 and CjLas1-Grc3 complexes, we conducted structural comparison by superposition the structures of the two complexes. Structural superposition shows that the conformations of all Grc3 domains and Las1 HEPN domains are almost identical in the two complexes, while the conformations of Las1 CC domains are significantly different (Figure 4A).

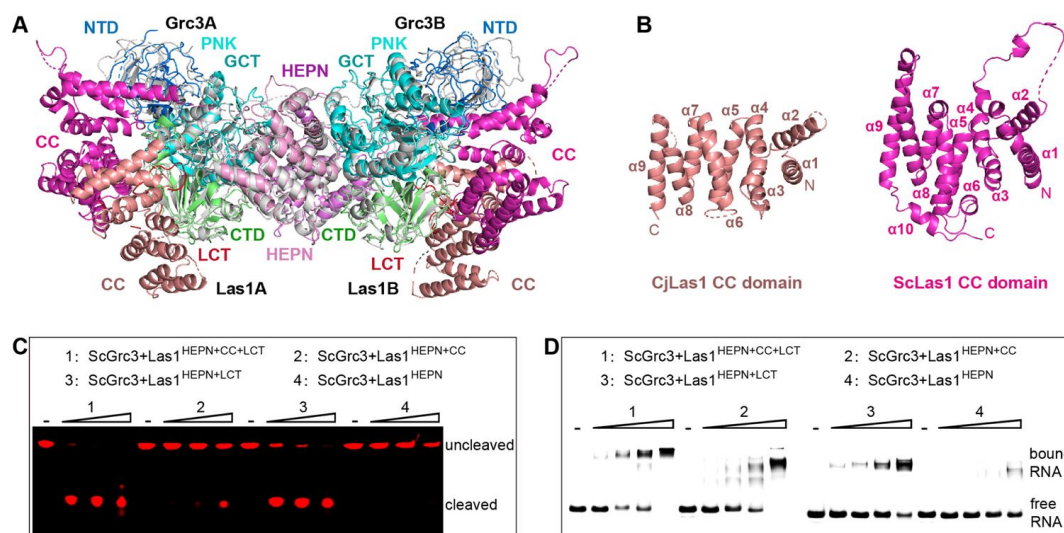


Figure 4.

The CC domain contributes to ITS2 RNA binding and cleavage.

(A) Structural comparison between ScLas1-Grc3 complex and CjLas1-Grc3 complex. Color-coding used for ScLas1 and ScGrc3 is identical to that used in Fig. 2a. The CC domain of CjLas1 is colored in salmon, other domains of CjLas1 and all domains of CjGrc3 are colored in gray. (B) Structures of CjLas1 CC domain (in salmon) and ScLas1 CC domain (in lightmagenta). (C) *In vitro* RNA cleavage assay using indicated truncations of ScLas1. (D) Electrophoretic mobility shift assay using indicated truncations of ScLas1.

Dali server reveals that CC domain shares little structural similarity with any known proteins (Holm & Rosenstrom, 2010). CC domain is mainly composed of α helices and is the largest domain of Las1 (Figure 4B), with low sequence similarity among different species (Figure 3-figure supplement 3), yet its function remains unknown. To determine whether it plays a role in Las1 catalyzing ITS2 cleavage, we performed EMSA and *in vitro* RNA cleavage assays using Las1 proteins with or without CC domain truncations. Our data shows that the CC domain contributes to the binding of ITS2 RNA and then facilitates ITS2 cleavage (Figure 4C, D), suggesting that the CC domain may play a role of ITS2 stabilization in the Las1 cutting reaction.

Grc3 GCT binds to the HEPN active center and mediates Las1 activation

The ScLas1-Grc3 tetramer structure shows that two Grc3 GCT are stabilized by two HEPN domains of Las1 (Figure 2). Each Grc3 GCT binds to a groove in each Las1 HEPN domain and extends to the ribonuclease active center (Figure 5A), which is formed by the two HEPN domains via dimerization. The conserved catalytic residues Arg129, His130, His134 with two copies from two HEPN domains form a symmetric catalytic active pocket (Figure 5B). The two C-terminals of Grc3 GCTs tightly bind to the catalytic active pocket through packing and hydrogen bond interactions. Specifically, the side chain of Trp617 within Grc3 GCT inserts into the active pocket and forms packing and hydrogen bond with catalytic residue Arg129 (Figure 5B). Moreover, Trp617 and His615 in Grc3 form extensive hydrogen bond with

residues Arg136, Leu99, Gly98 and His54 within Las1. The Trp617 residue is highly conserved in Grc3 (Figure 5C), and its mutation completely abolishes the ITS2 cleavage (Figure 5D), but has little effect on Las1 binding (Figure 5E), revealing that it plays a crucial role in the activation of HEPN endonuclease. We also mutated each of the residues at the C-terminal of Grc3 and examined the nuclease activity of Las1 complexes with these mutants. Our data shows that alanine substitution of multiply conserved residues dramatically reduces the ITS2 cleavage (Figure 5F, Figure 5—figure supplement 1), indicating that these residues are also essential for coordinating Las1 HEPN endonuclease activation.

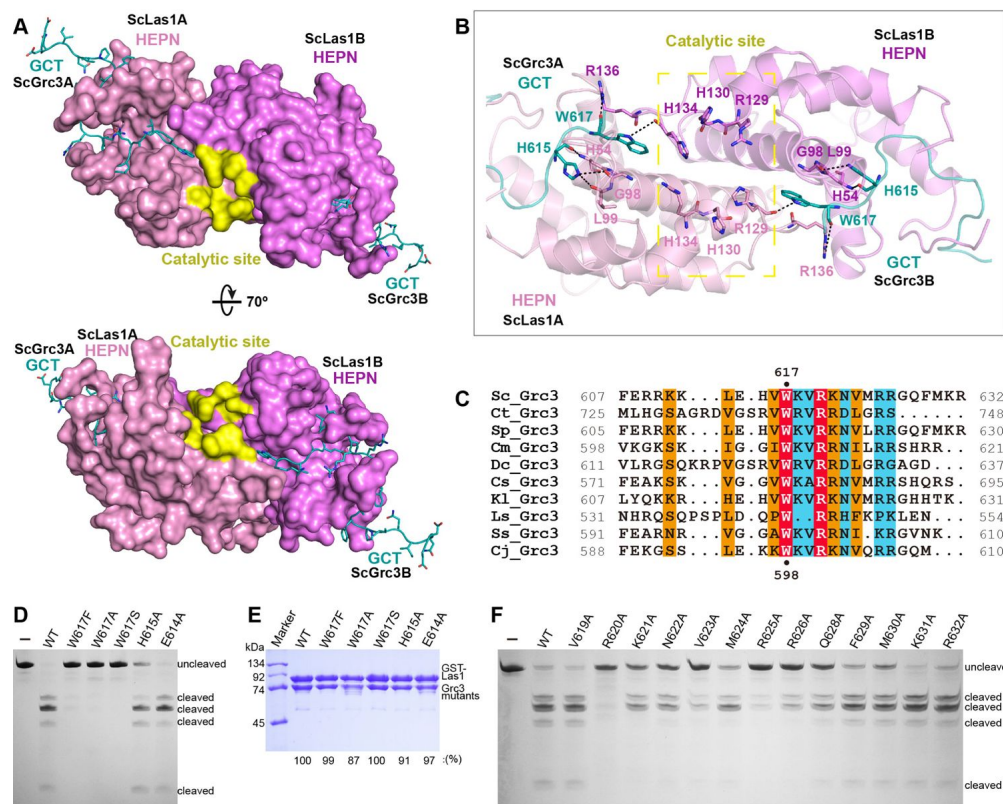


Figure 5.
ScGrc3 GCT mediates the ITS2 cleavage activity of ScLas1.

(A) ScGrc3 GCT binds at an active channel of ScLas1 HEPN dimer. Two HEPN domains of Las1 are colored in pink and violet, respectively. GCTs of Grc3 are colored in teal. The catalytic site is highlighted in yellow. HEPN domains are shown as surfaces, while GCTs are shown as sticks. (B) Detailed interactions between ScGrc3 GCT and ScLas1 HEPN domain. (C) Sequence alignments of Grc3 GCTs. Identical residues are highlighted in red. Basically constant

residuals are shaded in blue. Conserved residues are shaded in orange. (D) *In vitro* enzymatic assay of mutations of ScGrc3 residues Glu614, His615 and Trp617. (E) GST pull-down experiment assaying the ability of ScGrc3 mutants to interact with ScLas1. (F) *In vitro* enzymatic assay of alanine mutations of ScGrc3 C-terminal residues. The following figure supplement is available for Figure 5:

Las1 LCT drives Las1-Grc3 complex assembly

Our structure exhibits that the Las1 LCT is located in a groove in the CTD domain of Grc3 (Figure 6A). Structure analysis reveals that the Las1 LCT is stabilized by the Grc3 CTD through extensive hydrophobic interactions and hydrogen bonding. The side chains of Trp488, Trp494 and Phe499 of Las1 LCT are inserted into the three hydrophobic core regions of Grc3 CTD and form stable hydrophobic interactions with multitudinous hydrophobic residues of Grc3 (Figure 6B, C). Additionally, the side chain of Asn487 and the main chains of Lys497 and Ser489 of Las1 LCT form multiple hydrogen bonds with the sides of Trp573, His543 and the main side of Gln468 of Grc3 (Figure 6B, C). Mutations and deletions of Las1 LCT reduce the enzyme activity of Las1 (Figure 6—figure supplement 1). LCT deletions also

significantly affect the association between Las1 and Grc3 (Figure 6—figure supplement 2). These results highlight the functional significance of the sequence-dependent recognition of Las1 LCT by Grc3 CTD.

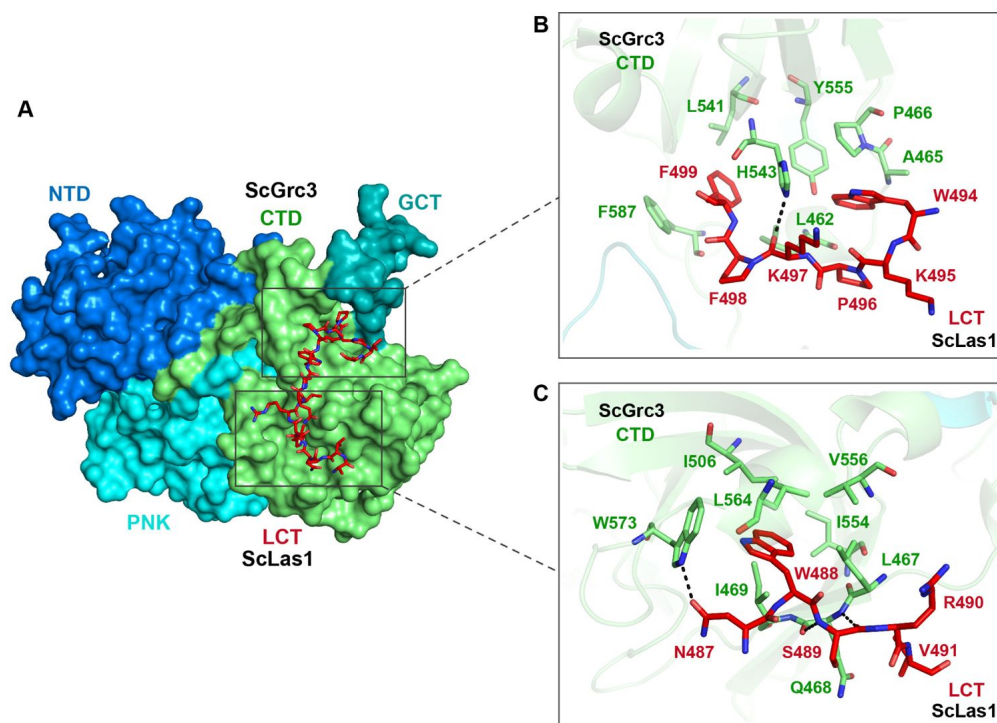


Figure 6.

Las1 LCT drives Las1-Grc3 cross talk.

(A) ScLas1 LCT binds to the CTD domain of ScGrc3. ScGrc3 is shown as surface, ScLas1 LCT is shown as stick. (B) Detailed interactions between C-terminal residues of ScLas1 LCT and ScGrc3 CTD domain. (C) Detailed interactions between N-terminal residues of ScLas1 LCT and ScGrc3 CTD domain. The following figure supplements are available for Figure 6:

Special crystal structure of Las1 HEPN domain

The biochemical data shows that Las1 exhibits very weak ability to cut ITS2 RNA at both C2 and C2' sites in the absence of Grc3 (Figure 1B, Figure 1—figure supplement 1), suggesting that Las1 may exist in a low-activity conformation state prior to assembly with Grc3 (Pillon et al., 2017). To determine the structure of Las1 in the low-activity conformation, we screened a lot of crystallization conditions with full-length Las1 proteins. Unfortunately, we did not obtain any crystals. We conjectured that the CC domain of Las1 might have a flexible conformation in the absence of Grc3, and then attempted to crystallize Las1 truncated by the CC domain and LCT. After screening a large number of crystallization conditions, we successfully obtained well-ordered crystals of the CjLas1 HEPN domain and determined its structure at 1.80 Å resolution using molecular replacement method (Figure 7A and Table 1).

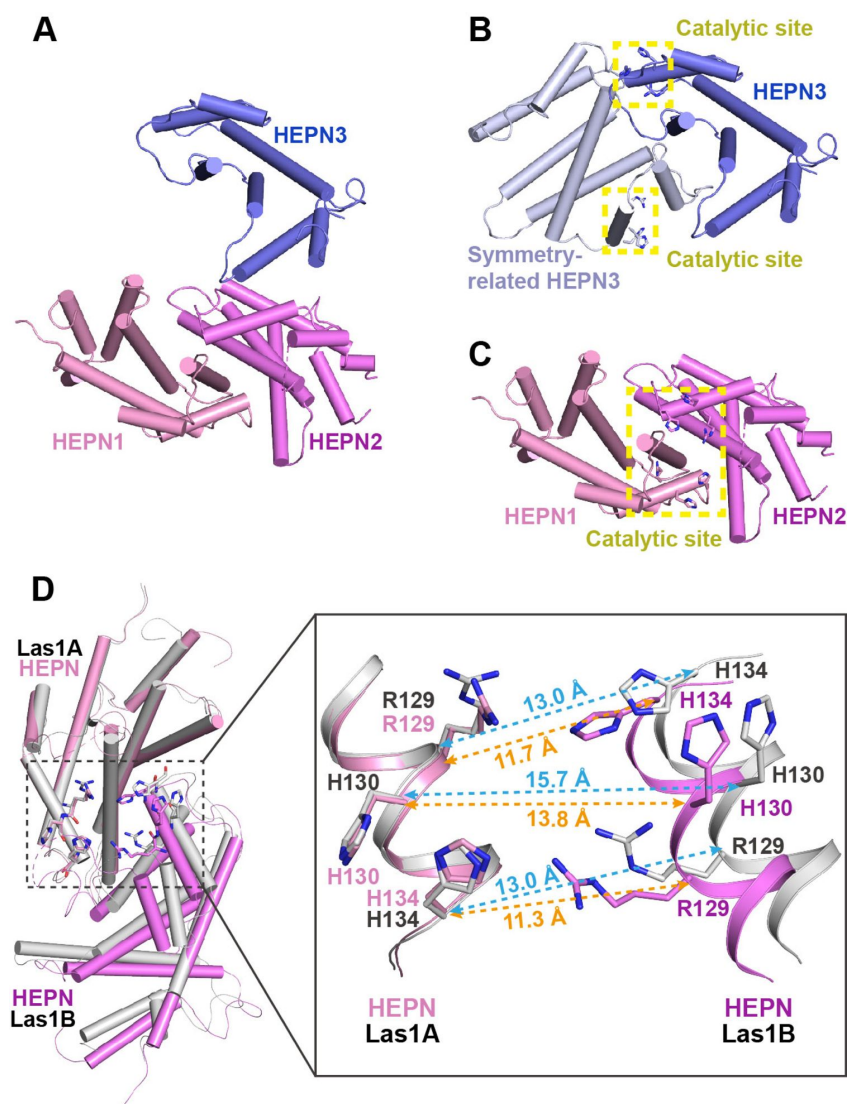


Figure 7.

Activation mechanism of Las1 by Grc3.

(A) Crystal structure of CjLas1 HEPN domain. (B) The HEPN3 molecule (in slate) and its symmetry-related molecule (in bluewhite) in Las1 HEPN domain structure. (C) The HEPN1 (in pink) and HEPN2 (in violet) molecules in Las1 HEPN domain structure. (D) Structural comparison of HEPN dimers between CjLas1-Grc3 complex (Las1A HEPN in pink, Las1B HEPN in violet) and CjLas1 HEPN domain (in gray). Inset: a magnified view of the comparison of the catalytic site in the two structures. The following figure supplements are available for Figure 7:

The crystallographic asymmetric unit contained three Las1 HEPN molecules (HEPN1, HEPN2 and HEPN3), each of which assumes an all α -helical fold (Figure 7A). Analysis of crystal packing and interactions across HEPN-HEPN interfaces suggests that the biological unit of Las1 HEPN domain may contain a homodimer and a monomer. HEPN1 and HEPN2 molecules form a face to face dimer related through the pseudo-2-fold axis, similar to the HEPN dimer in the Las1-Grc3 tetramer structure. Most notably, the molecule HEPN3 is likely not to form a dimer similar to HEPN1-HEPN2 with its symmetry-related molecule. Firstly, the structural superposition of HEPN3 and its symmetry-related molecule with the HEPN1-HEPN2 dimer shows that HEPN1 and HEPN3 molecules superimpose well with core root mean-square deviation (RMSD) of 0.7 Å for 133 Ca atoms, while HEPN2 and symmetry-related HEPN3 molecule exhibit significantly different conformations (Figure 7—figure supplement 1). Secondly, the catalytic residues in HEPN1 and HEPN2 domains are close to each other to form a compact active center, whereas the catalytic residues in HEPN3 and its symmetry-related molecule remain far apart, presenting two separate catalytic sites (Figure 7B, C). These observations suggest that the HEPN3 molecule in the asymmetric unit is probably to be a monomer, which forms a special packing with the dimer of HEPN1-HEPN2 in the crystal. Together, these results indicate a monomer-dimer equilibrium in the Las1 HEPN domain, which is consistent with the previously reported SEC-MALS data (Pillon et al., 2017).

Conformational changes in Las1 HEPN domain upon Grc3 binding

Since Grc3 binding significantly activates the endonuclease activity of Las1, the HEPN nuclease domains are probably to undergo conformational changes before and after Grc3 binding. We compared the HEPN domain dimer structures in CjLas1 and CjLas1-Grc3 complex by superposition and observed that significant conformational changes occur in HEPN domains upon Grc3 binding (Figure 7D). One HEPN domain is well aligned between the two structures, while the other HEPN domain has a large displacement. In order to investigate whether conformational changes occurred in the catalytic center, we further compared the catalytic pocket within the HEPN domains of the two structures. Remarkably, large conformational changes are observed in the catalytic pocket (Figure 7D). In the Grc3-free Las1 structure, the catalytic residues Arg129, His130 and His134 within one HEPN domain are far from the catalytic residues in the other HEPN domain, while in the Grc3-bound structure, the catalytic residues within two HEPN domains are close to each other (Figure 7D, Figure 7—figure supplement 2). These observations indicate that the Grc3 binding not only stabilizes the Las1 HEPN dimer, but also promotes the formation of a more compact catalytic pocket, showing better catalytic activity for specific cleavage of ITS2. Notably, compared with CjGrc3-bound CjLas1, ScGrc3-bound ScLas1 has a more compact catalytic pocket (Figure 7D, Figure 7—figure supplement 2), which may explain why ScLas1 shows better ITS2 cleavage activity than CjLas1 in the presence of Grc3 (Figure 1B, Figure 1—figure supplement 1).

Discussion

Removal of the ITS2 is the requirement for the maturation of 5.8S rRNA and 25S rRNA, which is an important step during the eukaryotic 60S subunit synthesis. In this study, the structural, biochemical, and functional analysis of ITS2 RNA processing machinery provides a critical step toward understanding the molecular mechanism of Grc3-activated ITS2 processing by Las1 endoribonuclease.

Mechanism of ITS2 processing by Las1-Grc3 dual enzyme complex

Based on our findings, we propose a model of ITS2 processing by Las1 endoribonuclease and its activator Grc3 kinase (Figure 8). In the absence of Grc3, Las1 has very weak ITS2 cleavage activity due to its unstable HEPN dimer. However, in the presence of Grc3, two copies of Las1 and Grc3 assemble into a stable tetramer that shows high activity to cleavage ITS2. Las1 initially processes ITS2 at the C2 site, generating a 5.8S rRNA precursor with 2',3'-cyclic phosphate and a 25S rRNA precursor with 5'-hydroxyl end (Figure 8 and Figure 11). The second step of ITS2 specific cleavage occurs at the C2' site to remove the 5'-region of the 25S rRNA precursor. The C2' cleavage also produces a 5'-hydroxyl product, which is rapidly phosphorylated by Grc3 only in the presence of ATP. After phosphorylation, the 25S precursor is further processed by a Rat1-Rai1 complex with 5'-3' exonuclease activity, which degrades the 5'-region of the 25S rRNA precursor and subsequently generate mature 25S rRNA (Figure 11) (Gasse et al., 2015). Furthermore, the nuclear exosome complex drives the maturation of the 5.8S RNA by removing the 3'-end of the precursor through its 3'-5' exonuclease activity (Fromm et al., 2017).

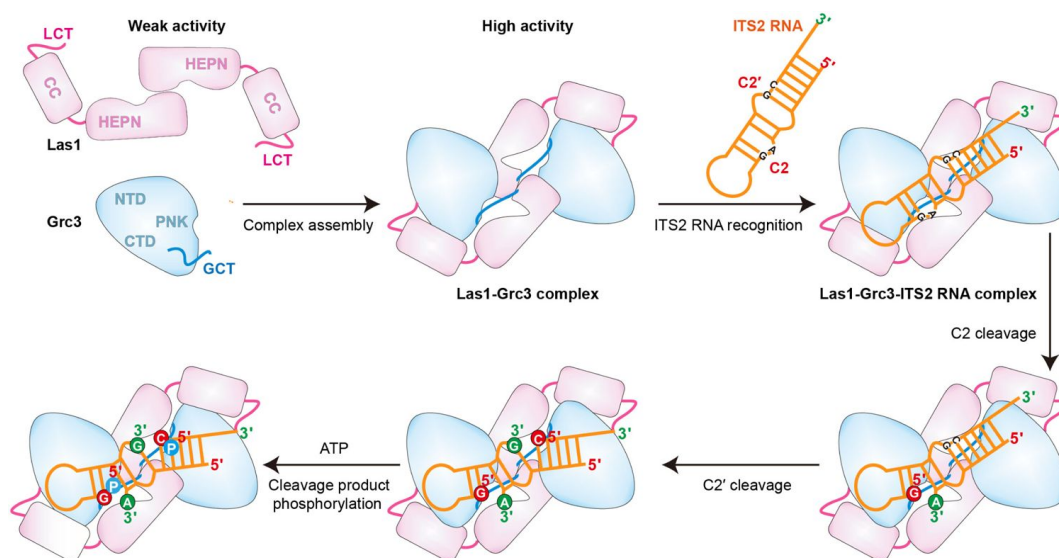


Figure 8.

Model for Grc3-mediated Las1-catalyzed ITS2 pre-rRNA processing.

Prior to assembly with Grc3, Las1 shows weak processing activity for ITS2 precursor RNA. When combined with Grc3 to form a tetramer complex, Las1 shows high processing activity for ITS2 precursor RNA. Las1 specifically cleaves ITS2 at C2 and C2' sites to generate 5'-OH terminus products. The 5'-OH terminus products are further phosphorylated by Grc3 when in the presence of ATP. The following figure supplement is available for [Figure 8](#):

Distinct activation mechanism for Las1 HEPN nuclease

Structural and biochemical results highlight the mechanism of Grc3 GCT-mediated Las1 HEPN nuclease activation, which is quite distinct from other HEPN nucleases such as CRISPR-Cas and toxin-antitoxin associated HEPN RNases, Cas13a, Csm6, RnIA. Though proper dimerization of the HEPN domain is critical for HEPN nuclease activity, other factors are also common requirements for HEPN activation. For examples, activation of the Cas13a HEPN enzyme requires target RNA binding and guide-target RNA duplex formation, while allosterically activation of the Csm6 HEPN nuclease is dependent on a cyclic oligoadenylate (Liu, Li, Ma, et al., 2017; (Niewoehner et al., 2017)). In addition, stimulating the toxicity of RnIA HEPN RNase requires an association between RnIA and RNase HI, and triggering activation of the HEPN domains of Ire1 and RNase L needs binding of ATP to their kinase domains (Lee et al., 2008; Naka, Koga, Yonesaki, & Otsuka, 2014; Wang et al., 2014). It is noteworthy that these activators do not interact directly with the HEPN domain, but bind to other domains to induce a conformational transition of the HEPN domain from inactive to active state. In contrast, Grc3 GCTs bind directly to the active center of HEPN domains and form hydrogen bonds with catalytic residues, which appears to provoke rearrangements of the active site required for activation of HEPN. Regulation of HEPN RNases by interacting with catalytic residues may be a direct and effective measure.

Comparison of Las1-Grc3 complex and other nuclease-kinase machines

Ire1 and RNase L are also nuclease-kinase machines that contain a HEPN domain and a protein kinase domain or a pseudo-protein kinase domain, playing a fundamental role in RNA degradation related to a variety of cellular processes. Although Ire1 and RNase L also require higher-order assembly during RNA degradation, the Las1-Grc3 complex shows distinct structure assembly with them ([Figure 8—figure supplement 1](#)). Both the kinase domain and the HEPN domain in Ire1 and RNase L adopt a parallel back to back dimer configuration ([Lee et al., 2008](#); [Wang et al., 2014](#)). In Las1, only the HEPN domain forms a dimer architecture, and Grc3 does not contact each other, but binds on both sides to stabilize the HEPN dimer. An ankyrin repeat domain in RNase L and an endoplasmic reticulum (ER) luminal domain in Ire1 has been proposed to promote dimerization of the kinase and HEPN domains ([Credle, Finer-Moore, Papa, Stroud, & Walter, 2005](#); [Huang et al., 2014](#)). Whereas the additional CC domain in Las1 is likely to contribute to RNA binding and facilitate cleavage ([Figure 4C, D](#)). In addition, Las1, Ire1, and RNase L recognize similar RNA cleavage motifs despite being involved in different RNA processing, splicing, and degradation pathways. Cleavage studies show a preference of Las1 for UAG and UGC motifs, which are also the universal cleavage motifs observed in mRNA decay by Ire1 and RNase L ([Pillon & Stanley, 2018](#)).

Materials and methods

Protein expression and purification

For Las1-Grc3 complex expression and purification, *Saccharomyces cerevisiae* Las1 (*ScLas1*) gene was cloned into pET23a vector with an N-terminal His₆-tag and *Cyberlindnera jadinii* (*CjLas1*) gene was cloned into pET28a vector with an N-terminal His₆SUMO-tag, while *ScGrc3* and *CjGrc3* genes were cloned into modified pET28a vector containing an N-terminal SUMO-tag, followed by an ubiquitin-like protein1 (Ulp1) protease cleavage site. All recombinant plasmids were transformed into *E.coli* Rosetta (DE3) (Novagen) cells and grown in LB broth at 37°C for 3 h. After culturing to an OD₆₀₀ of 0.6-0.8, protein expression was induced with 0.2 mM isopropyl-1-thio-β-D-galactopyranoside (IPTG) at 16°C for 14 h. For the purification of ScLas1-Grc3 complex proteins, cells expressing ScLas1 proteins and ScGrc3 proteins were collected and co-lysed by sonication in buffer containing 20 mM Tris-HCl, pH 7.5, 500 mM NaCl. After centrifugation, supernatant was incubated with Ni Sepharose (GE Healthcare), and the bound protein was eluted with buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 300 mM imidazole. Eluted protein was digested with Ulp1 protease at 4°C for 2 h and then further purified on a Heparin HP column (GE Healthcare), eluting with a linear gradient of increasing NaCl concentration from 300 mM to 1 M in 20 mM Tris-HCl, pH 7.5 buffer. The fractions containing the protein of interest were concentrated and further purified by size-exclusion chromatography (Superdex 200 Increase 10/300, GE Healthcare) in a buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, and 1 mM Tris(2-carboxyethyl)phosphine hydrochloride (TCEP). Proteins were collected and concentrated to a final concentration of 15 mg/ml. For the purification of CjLas1-Grc3 complex proteins, cells expressing CjLas1 proteins and CjGrc3 proteins were collected and co-lysed by sonication in buffer containing 20 mM Tris-HCl, pH 7.5, 500 mM NaCl. After centrifugation, supernatant was incubated with Ni Sepharose (GE Healthcare), and the bound protein was eluted with buffer containing 20 mM Tris-HCl, pH 7.5, 350 mM NaCl, 300 mM imidazole. Eluted protein was digested with Ulp1 protease at 4°C for 2 h and then further purified on a Heparin HP

column (GE Healthcare), eluting with buffer containing 20 mM Tris-HCl, pH 7.5, 1M NaCl. The protein was further purified by size-exclusion chromatography (Superdex 200 Increase 10/300, GE Healthcare) in a buffer containing 20 mM Tris-HCl, pH 7.5, 350 mM NaCl, and 1 mM Tris(2-carboxyethyl)phosphine hydrochloride (TCEP). Proteins were collected and concentrated to a final concentration of 15 mg/ml. Mutants were expressed and purified as wild-type protein. The CjLas1 HEPN construct was purified with an identical protocol.

For the experiment of GST pull-down, the wild-type *ScLas1* gene and its mutants were cloned into a modified pET23a vector (Novagen) with an N-terminal His₆-GST-tag, while the wild-type *ScGrc3* gene and its mutants were cloned into a modified pET28a vector containing an N-terminal His₆-SUMO-tag. Proteins were overexpressed in *E. coli* Rosetta (DE3) (Novagen) cells and cultured in LB broth to an OD₆₀₀ of 0.6-0.8. Then the target proteins were induced with 0.2 mM isopropyl-1-thio- β -D-galactopyranoside (IPTG) and grown for additional 14 h at 16°C. Cells were collected and lysed by sonication in buffer containing 20 mM Tris-HCl, pH 7.5, 500 mM NaCl. After centrifugation, supernatant was incubated with Ni Sepharose (GE Healthcare), and the bound protein was eluted with buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 300 mM imidazole. The GST-ScLas1 mutants were further purified on a Heparin HP column (GE Healthcare) and ScGrc3 mutants were purified on a HitTrap Q HP column (GE Healthcare). All proteins were further purified by size-exclusion chromatography (Superdex 200 Increase 10/300, GE Healthcare) in a buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl.

Crystallization, data collection and structure determination

Crystals of ScLas1-Grc3 complex, CjLas1-Grc3 complex and CjLas1 truncated protein (HEPN domain) were first obtained using the sitting drop vapor diffusion method using high-throughput crystallization screening kits (Hampton Research, Molecular Dimensions and QIAGEN). Crystals were then grown in a mixed solution containing 1 μ l complex solution and 1 μ l of reservoir solution using the hanging drop vapor diffusion method at 16°C. For growing large crystals, crystals were further optimized by using seeding technique. Well-diffracting crystals of ScLas1-Grc3 complex were grown in a reservoir solution containing 2% (v/v) tacsimate, pH 7.0, 0.1 M imidazole, pH 7.0, 2% (v/v) 2-propanol, and 9% (w/v) PEG 3350. Well-diffracting crystals of CjLas1-Grc3 complex were grown in a reservoir solution containing 0.1 M sodium phosphate, pH 7.5, 0.05 M NaCl, and 9% (w/v) PEG 4000. The best crystals of CjLas1 HEPN domain protein were grown from 0.1 M Tris-HCl, pH 8.0, 0.2 M MgCl₂, and 25% (w/v) PEG 3350. All crystals soaked in cryoprotectants made from the mother liquors supplemented with 20% (v/v) glycerol and flash frozen in liquid nitrogen.

All diffraction data sets were collected at beamline BL-17U1, BL-18U1, and BL-19U1 at Shanghai Synchrotron Radiation Facility (SSRF) and National Center for Protein Sciences Shanghai (NCPSS), and processed with HKL3000 (Otwinowski & Minor, 1997). The CjLas1 HEPN structure were determined by molecular replacement using the HEPN domain structure within CtLas1-Grc3 (PDB: 6OF4) as the search model using the program PHENIX Phaser (Adams et al., 2002; Pillon et al., 2019). The phases of ScLas1-Grc3 complex and CjLas1-Grc3 complex were solved by molecular replacement method with the cryo-EM maps of Las1-Grc3 complexes using PHENIX Phaser. The model was manually built and adjusted using the program COOT (Emsley, Lohkamp, Scott, & Cowtan, 2010). Iterative cycles of crystallographic refinement were performed using PHENIX. All data processing and structure refinement statistics are summarized in Supplementary Table 1. Structure figures were prepared using PyMOL (<http://www.pymol.org/>).

Cryo-EM data acquisition

The samples were diluted at a final concentration of around 1.0 mg/mL. Three microliters of the samples were applied onto glow-discharged 200-mesh R2/1 Quantifoil copper grids. The grids were blotted for 4 s and rapidly cryocooled in liquid ethane using a Vitrobot Mark IV (Thermo Fisher Scientific) at 4°C and 100% humidity. The samples were imaged in a Titan Krios cryo-electron microscope (Thermo Fisher Scientific) at a magnification of 105,000 $\times$ (corresponding to a calibrated sampling of 0.82 Å per pixel). Micrographs were recorded by EPU software (Thermo Fisher Scientific) with a K3 detector, where each image was composed of 30 individual frames an exposure time of 3 s and an exposure rate of 16.7 electrons per second per Å². A total of 2,520 movie stacks for ScLas1-Grc3 complex and 8,616 movie stacks for CjLas1-Grc3 complex were collected.

Single-particle image processing and 3D reconstruction

All micrographs were first imported into Relion (Scheres, 2012) for image processing. The motion-correction was performed using MotionCor2 (Zheng et al., 2017) and the contrast transfer function (CTF) was determined using CTFFIND4 (Rohou & Grigorieff, 2015). All particles were autopicked using the NeuralNet option in EMAN2 (Tang et al., 2007). Then, particle coordinates were imported to Relion, where the poor 2D class averages were removed by several rounds of 2D classification. Initial maps were built and classified using ab-initio 3D reconstruction in cryoSPARC (Punjani, Rubinstein, Fleet, & Brubaker, 2017) without any symmetry applied. Heterogeneous refinement was further performed to remove bad particles using one good and one bad starting maps. The good class having 264,341 particles for ScLas1-Grc3 complex or 523,843 particles for CjLas1-Grc3 complex was selected and subjected to 3D homogeneous refinement, local & global CTF refinement, and non-uniform refinement with C2 symmetry imposed, achieving a 3.07 Å resolution map for the ScLas1-Grc3 complex and a 3.39 Å resolution map for the CjLas1-Grc3 complex, respectively. Resolution for the final maps were estimated with the 0.143 criterion of the Fourier shell correlation curve. Resolution maps were calculated in cryoSPARC using the “Local Resolution Estimation” option. (More information in Supplementary Figure S2, Supplementary Figure S3, and Supplementary Table 2).

In vitro transcription of RNA

Unlabeled ITS2 RNAs used for cleavage assays were synthesized by *in vitro* transcription with T7 RNA polymerase and linearized plasmid DNAs as templates. Transcription reactions were performed at 37°C for 4 h in buffer containing 100 mM HEPES-KOH, pH 7.9, 20 mM MgCl₂, 30 mM DTT, 2 mM each NTP, 2 mM spermidine, 0.1 mg/ml T7 RNA polymerase, and 40 ng/μl linearized plasmid DNA template. The transcribed RNA was then purified by gel electrophoresis on a 12% denaturing (8 M urea) polyacrylamide gel, RNA band was excised from the gel and recovered with Elutrap System. The purified RNA was resuspended in DEPC (diethyl pyrocarbonate)-treated water.

In vitro ITS2 RNA cleavage assays

For 5'-Cy5 and 3'-Cy3 labeled ITS2 RNA cleavage assays, 0.5 μM of ITS2 RNA was incubated at 37°C for 2 h with increasing amount of Las1, Grc3 or Las1-Grc3 complex proteins (0.05 μM–0.5 μM) in cleavage buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 1 mM Tris(2-carboxyethyl)phosphine hydrochloride (TCEP). 5'-Cy5 and 3'-Cy3 labeled ITS2 RNA was synthesized from Takara Biomedical Technology. Reactions were stopped by adding 2 $\times$ loading buffer. Samples were analyzed on a 20% urea denaturing polyacrylamide gel with

TBE buffer. Cleavage products were visualized by fluorescent imaging and analysis system (SINSAGE TECHNOLOGY).

For unlabeled ITS2 RNA cleavage assays, 10 μ M of ITS2 RNA was incubated at 37°C for 2 h with increasing amount of Las1, Grc3 or Las1-Grc3 complex proteins (1.5 μ M-15 μ M) in cleavage buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 1 mM Tris(2-carboxyethyl)phosphine hydrochloride (TCEP). Reactions were stopped by adding 2 $\times$ loading buffer and were then quenched at 75°C for 5 min. Samples were analyzed on a 20% urea denaturing polyacrylamide gel with TBE buffer. Cleavage products were visualized by toluidine blue staining.

RNA cleavage products phosphorylation assays

10 μ M of unlabeled ITS2 RNA was incubated at 37°C for 1.5 h with 10 μ M of Las1-Grc3 complex proteins in cleavage buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 1 mM Tris(2-carboxyethyl)phosphine hydrochloride (TCEP). 1 mM ATP was added into each reaction system at 37°C for 30 min. Reactions were stopped by adding 2 $\times$ loading buffer and were then quenched at 75°C for 5 min. Samples were analyzed on a 20% urea denaturing polyacrylamide gel with TBE buffer. Cleavage products were visualized by toluidine blue staining.

GST pull-down assays

Pull-down experiments were carried out using GST fusion proteins to analyze the association between Las1 and Grc3. 0.2 mg purified GST-ScLas1 protein were incubated with 0.4 mg purified ScGrc3 protein in binding buffer containing 20 mM Tris-HCl, pH 7.5, 500 mM NaCl, 2 mM DTT. 80 μ l GST affinity resin was added into each reaction system at 4°C for 60 min. The resin was washed with 1 ml of binding buffer. After washing five times, the binding samples were eluted with elution buffer containing 20 mM Tris, pH 7.5, 20 mM GSH, 500 mM NaCl, and 2 mM DTT. The elution samples were then monitored using SDS-PAGE (polyacrylamide gel electrophoresis) and visualized by Coomassie blue staining. The assays were quantified by band densitometry. The experiment was repeated three times.

Electrophoretic mobility shift assays

Electrophoretic mobility shift assays (EMSA) were performed with a series of Las1-Grc3 complex dilutions from 20 μ M to 2 μ M and a 81-nt ITS2 RNA. The ITS2 RNA was synthesized by in vitro transcription. Proteins were incubated with ITS2 RNA in a binding buffer containing 20 mM Tris-HCl, pH 7.5, 300 mM NaCl for 30 min at 4°C. After reaction, the binding samples were then resolved on 5% native acrylamide gels in Tris-Glycine (0.5X TBE) buffer pH 8.5 under an electric field of 100 volts for 40 min at 4°C. Gels were imaged by using a ChemiDoc XRS+ (Bio-rad).

RNA sequencing

Enzyme-digested products from the ITS2 cleavage assay were isolated by dialysis from denaturing polyacrylamide gel. T4 PNK was used to phosphorylate the 5' terminus of RNA products. After the above treatment, RNA samples were linked with the adaptor RNA by T4 RNA ligase-1. RNA samples were then treated for reverse transcription and PCR amplification. The PCR amplification products were cloned into pET28a vector, then sequenced.

Data availability

The atomic coordinates included in this study have been deposited in the Protein Data Bank (PDB) with the following accession codes: 7Y16, 7Y17, and 7Y18. Cryo-EM maps of the ScLas1-Grc3 complex and CjLas1-Grc3 complex in this study have been deposited in the wwPDB OneDep System under EMD accession code EMD-33733 and EMD-33735.

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Author contributions

J.C., H.C., X.L., and R.H. expressed and purified the proteins and grew crystals. J.C., H.C., and L.L. collected X-ray diffraction data. S.L., K.Z., and L.L. solved the structures. X.L., H.C., R.H., and J.C. carried out all of the cloning and performed the biochemical assays. L.L. prepared the figures; L.L. wrote the manuscript and supervised all of the research.

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- Liang Liu

The funders had no role in study design, data collection and interpretation, or the decision to submit the work for publication.

Competing interests

The authors declare no competing interests.

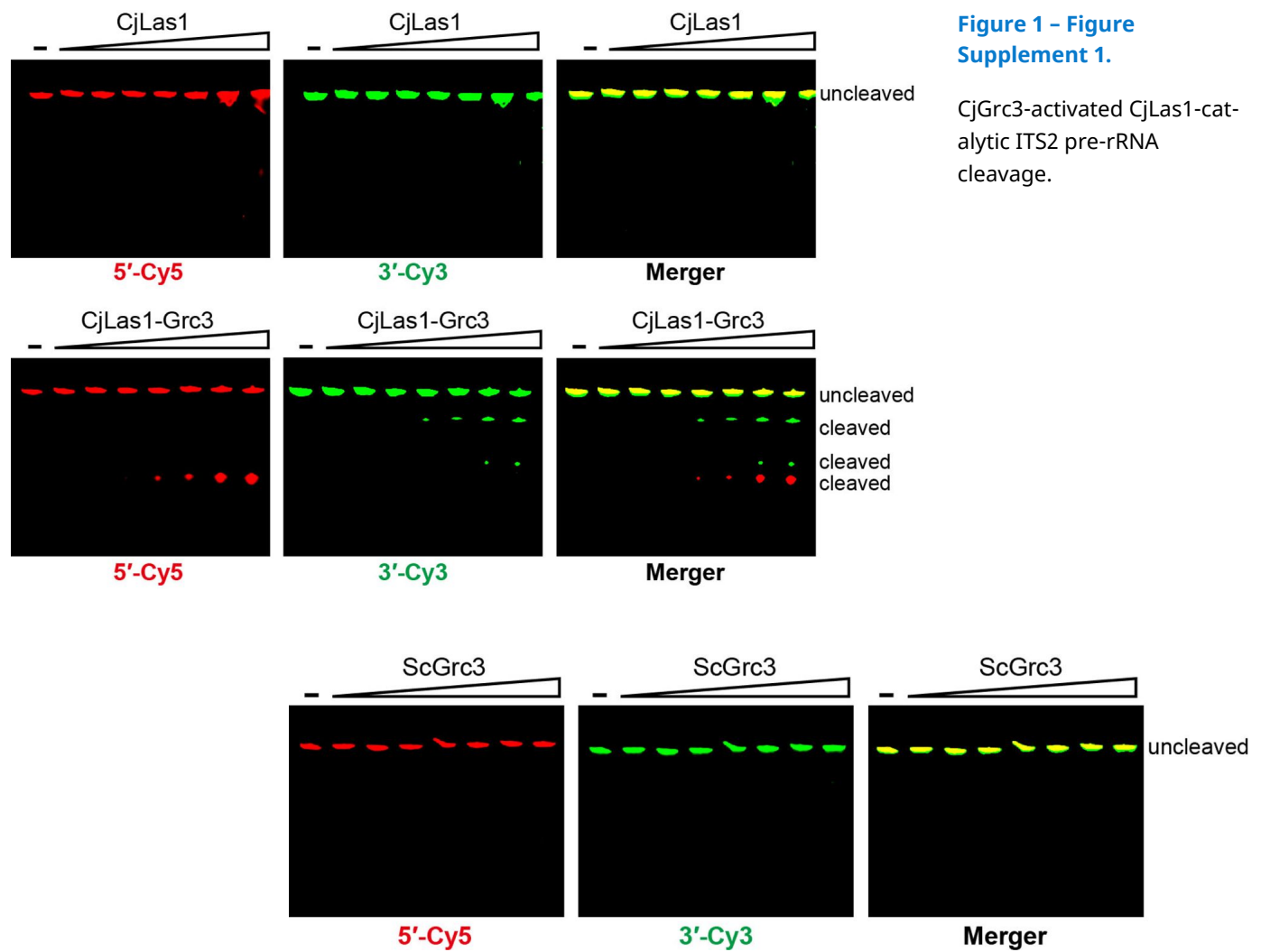
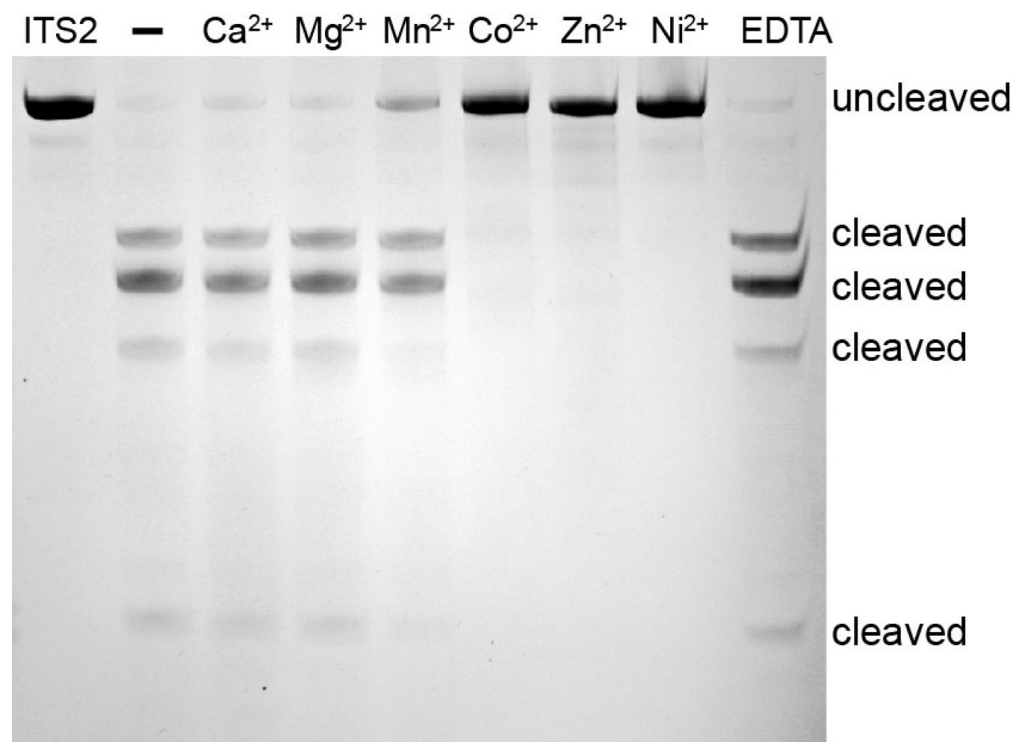
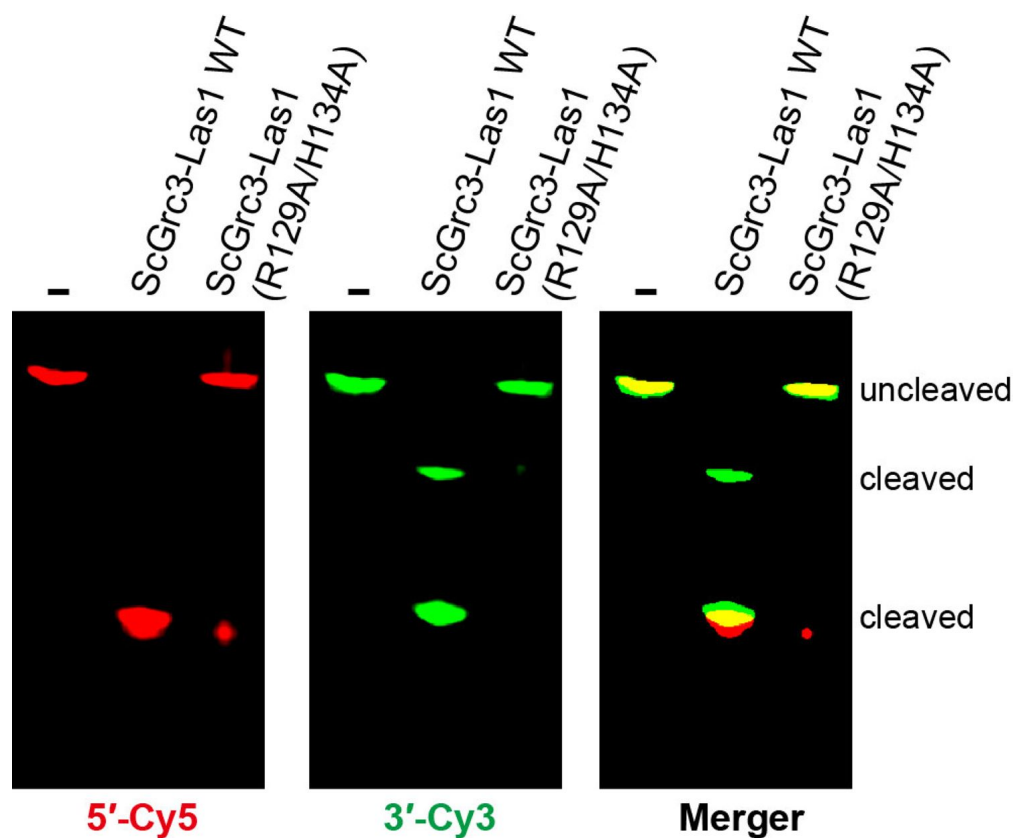


Figure 1 - Figure Supplement 2.

ScGrc3 has no ITS2 pre-rRNA cleavage activity.



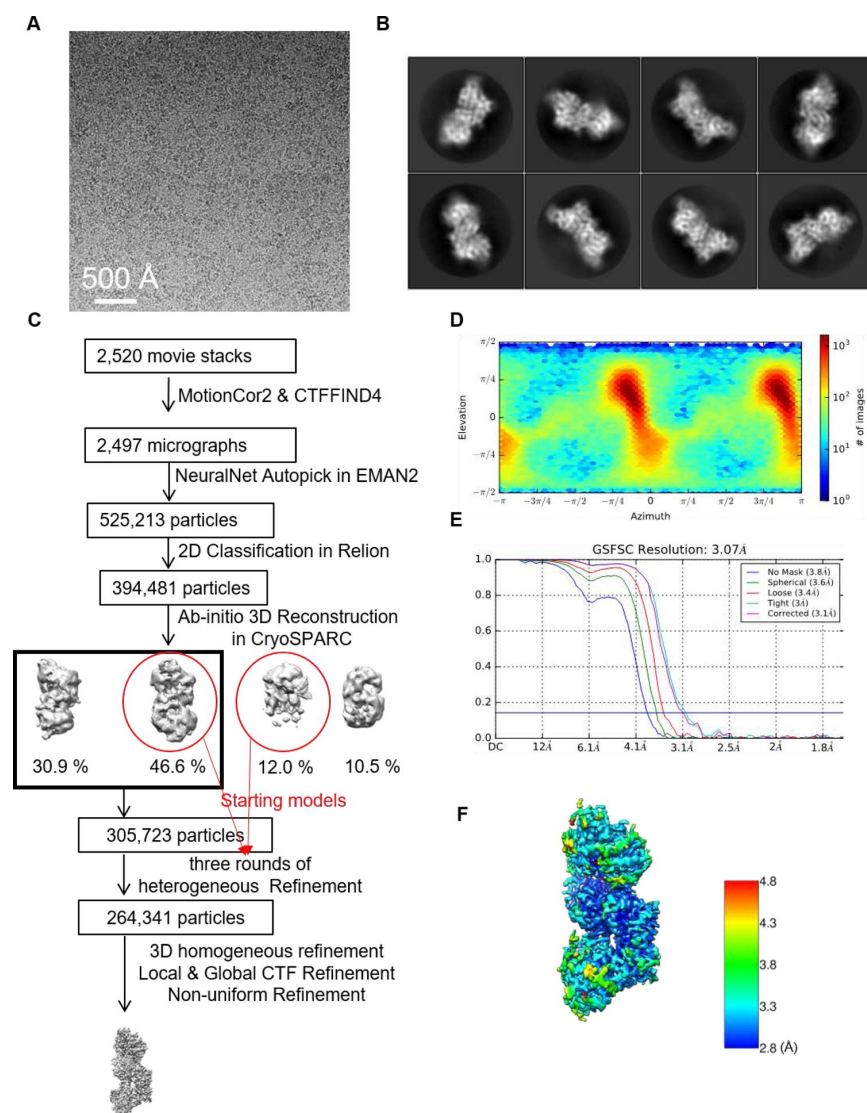


Figure 2 – Figure Supplement 1.

Single-particle cryo-EM analysis of the ScLas1-Grc3 complex.

(A) Representative motion-corrected cryo-EM micrograph. **(B)** Reference-free 2D class averages. **(C)** Workflow of the data processing. **(D)** Euler angle distribution of the particle images. **(E)** Gold standard FSC plot for the 3D reconstruction with full particle set, calculated in cryoSPARC. **(F)** Resolution map for the final 3D reconstruction.

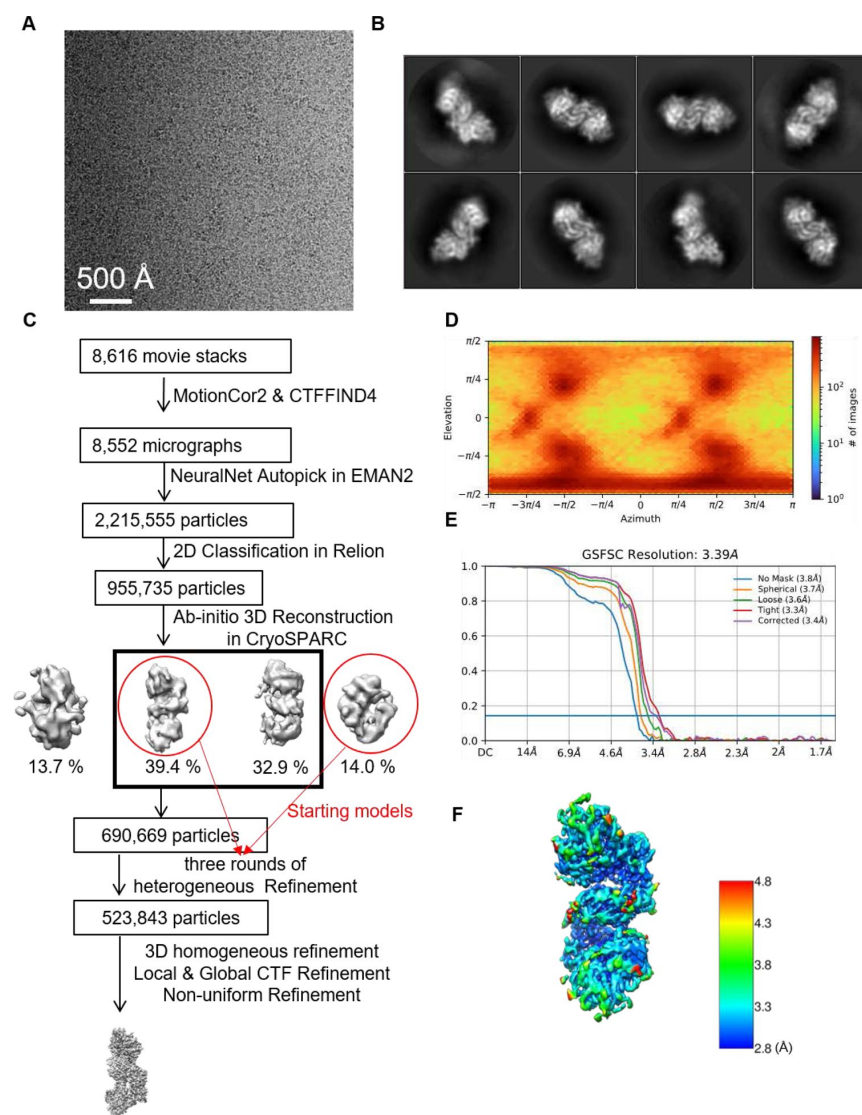


Figure 3 – Figure Supplement 1.

Single-particle cryo-EM analysis of the CjLas1-Grc3 complex.

(A) Representative motion-corrected cryo-EM micrograph. (B) Reference-free 2D class averages. (C) Workflow of the data processing. (D) Euler angle distribution of the particle images. (E) Gold standard FSC plot for the 3D reconstruction with full particle set, calculated in cryo-SPARC. (F) Resolution map for the final 3D reconstruction.

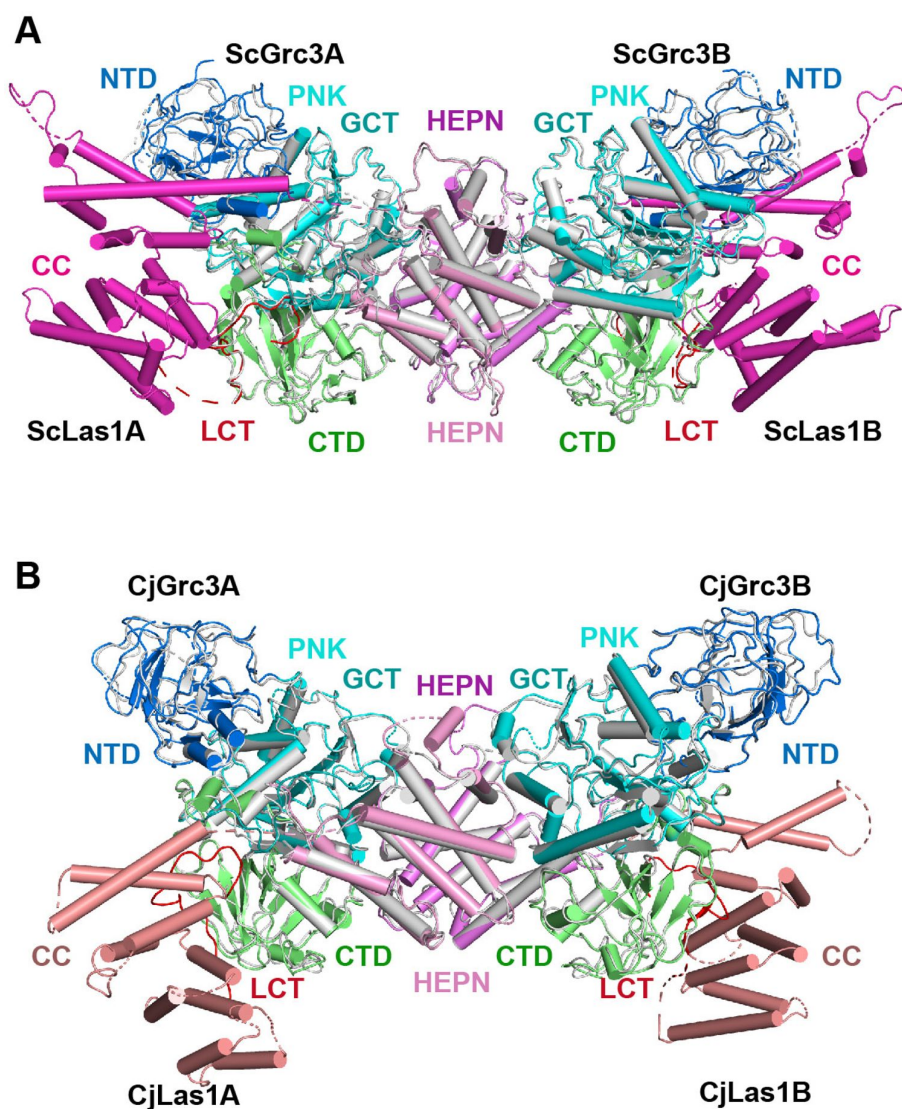


Figure 3 – Figure Supplement 2.

Comparison of cryo-EM and crystal structures of Las1-Grc3 complexes.

(A) Superposition of the ScLas1-Grc3 complex crystal structure with the cryo-EM structure. Color-coding used for ScLas1 and ScGrc3 in crystal structure is identical to that used in Fig. 2a. ScLas1 and ScGrc3 in cryo-EM structure are colored in gray. (B) Superposition of the CjLas1-Grc3 complex crystal structure with the cryo-EM structure. Color-coding used for CjLas1 and CjGrc3 in crystal structure is identical to that used in Fig. 3a. CjLas1 and CjGrc3 in cryo-EM structure are colored in gray.

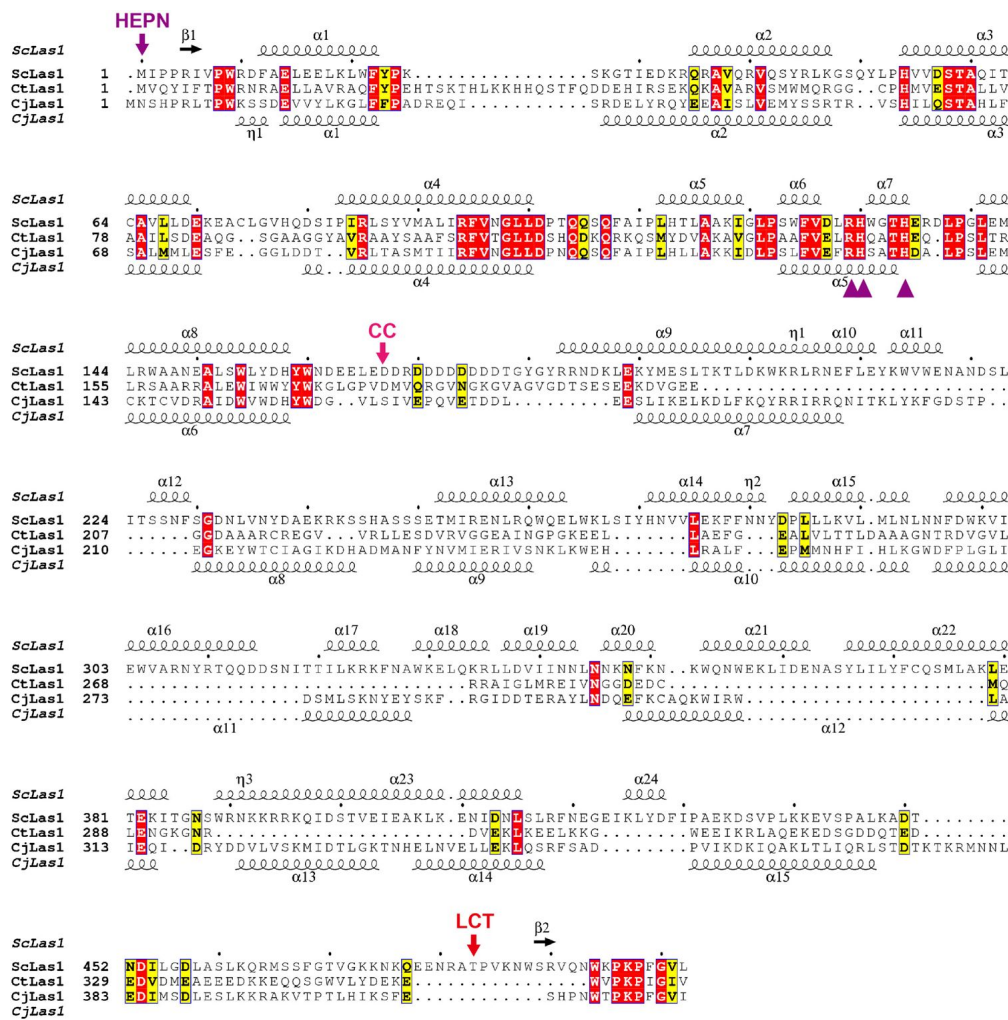


Figure 3 – Figure Supplement 3.

Sequence alignment of Las1 proteins.

The ScLas1 protein was aligned with its respective homologs. Alignment was performed using the MUSCLE program. The secondary structures of ScLas1 and CjLas1 are shown above and below the sequences, respectively. Conserved residues are highlighted in yellow, and invariant residues are highlighted in red. Catalytic residues in Las1 HEPN domain are marked in violet triangles. The figure was prepared using ESPrpt3.

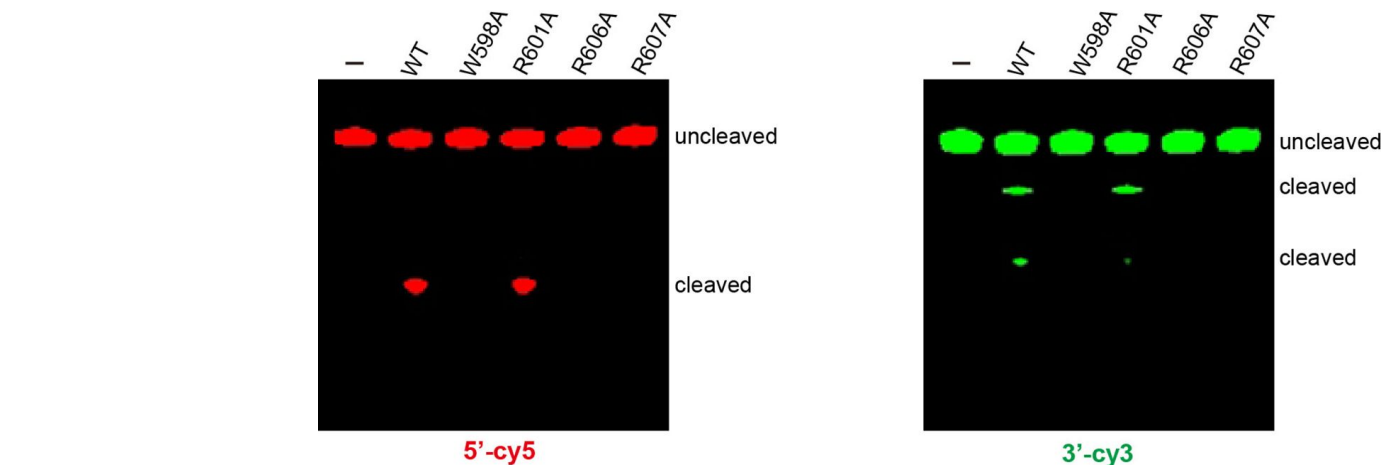


Figure 5 – Figure Supplement 1.

In vitro enzymatic assay of alanine mutations of conserved CjGrc3 residues Trp618, Arg601, Arg606 and Arg607.

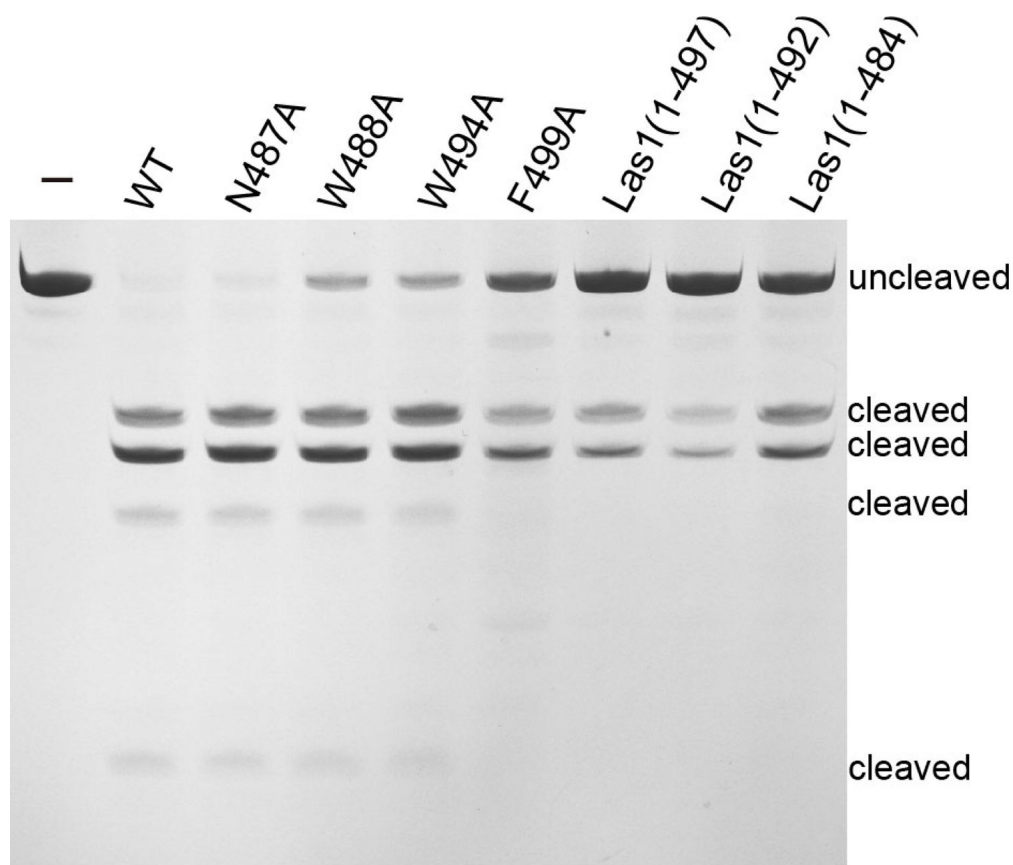


Figure 6 – Figure Supplement 1.

Denaturing gel showing the ITS2 pre-RNA cleavage by mutation or deletion of the interacting residues of ScLas1 LCT.

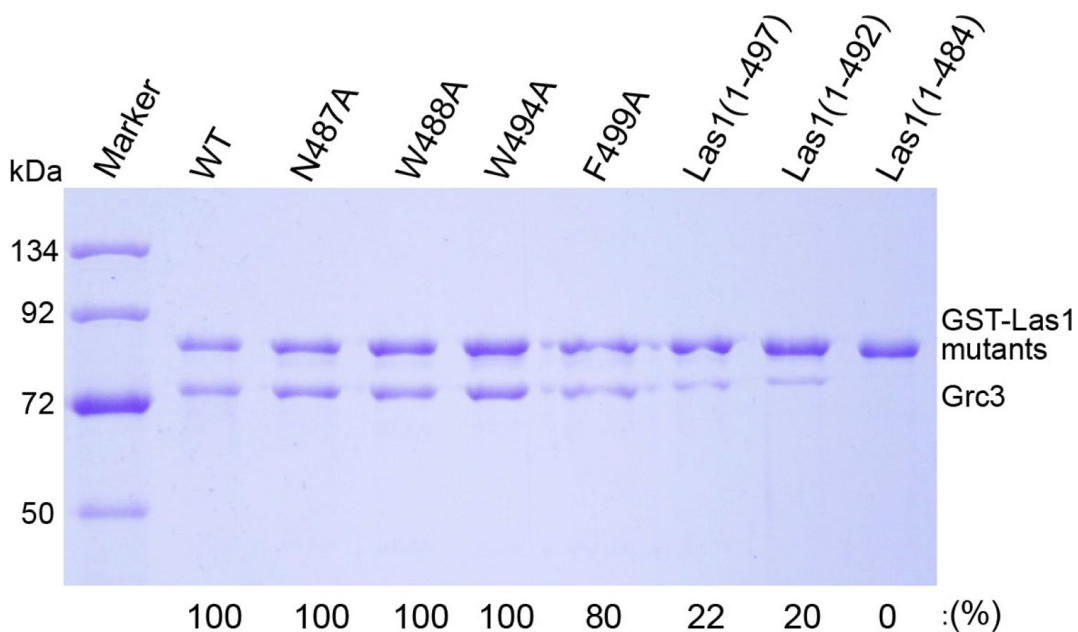


Figure 6 – Figure Supplement 2.

GST pull-down experiment assaying the Grc3 binding ability by mutation or deletion of the interacting residues of ScLas1 LCT.

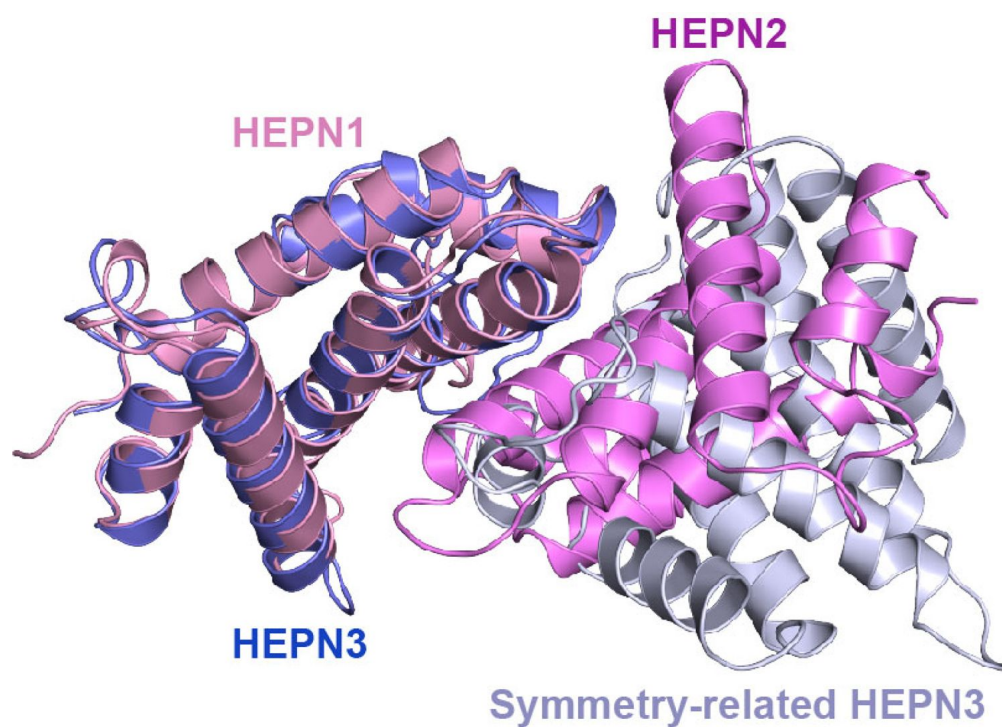


Figure 7 – Figure Supplement 1.

Structural superposition of HEPN3 (in slate) and its symmetry-related molecule (in bluewhite) with the HEPN1(in pink)-HEPN2 (in violet) dimer.

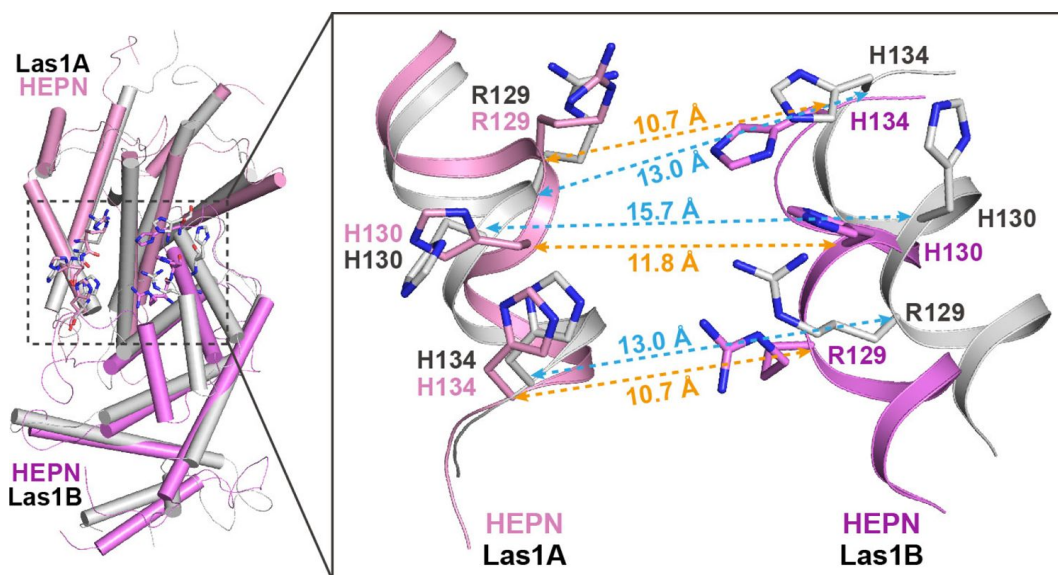


Figure 7 – Figure Supplement 2.

Structural comparison of HEPN dimers between ScLas1-Grc3 complex (Las1A HEPN in pink, Las1B HEPN in violet) and CjLas1 HEPN domain (in gray). Inset: a magnified view of the comparison of the catalytic site in the two structures.

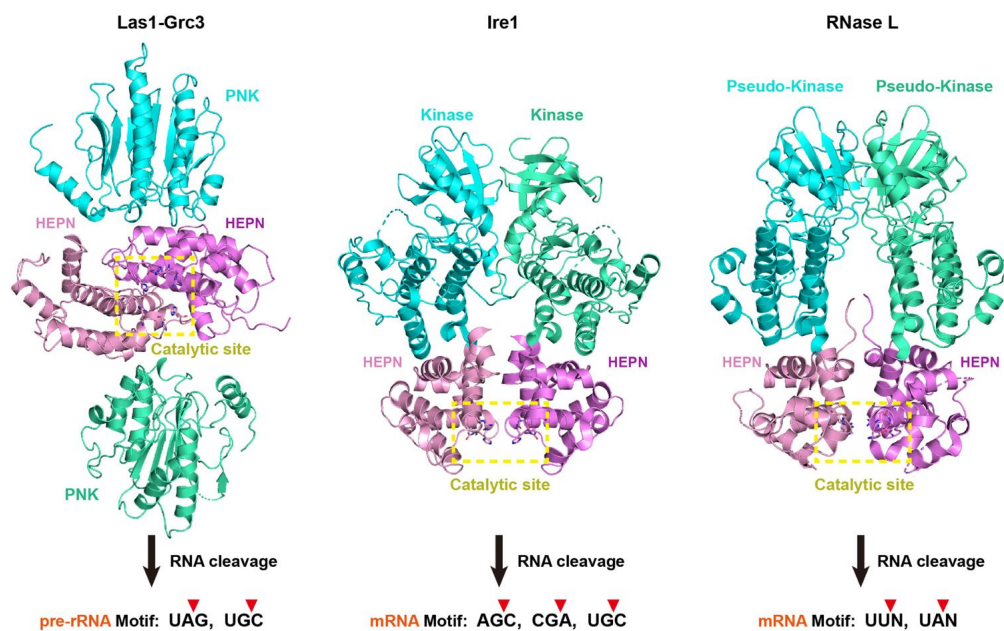


Figure 8 – Figure Supplement 1.

Comparison of Las1-Grc3, Ire1 and RNase L.

Comparison of structural assembly, activation and substrate recognition mechanisms of Las1-Grc3, Ire1 and RNase L. Two HEPN domains in Las1, Ire1 and RNase L are colored in pink and violet, respectively. Two PNK domains in Grc3, two Kinase domains in Ire1, and two Pseudo-Kinase domains in RNase L, are colored in cyan and greencyan, respectively.

tively. The catalytic site is highlighted with a yellow box.

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Reviewer #1 (Public Review):

In the manuscript there is not much comparison between the crystal and cryoEM structures provided, and on inspection they appear to be very similar. The crystal structures also reveal parts of the CC domains in Las1, which is not present in the cryoEM structures. It is interesting the CC domains in Sc and Cj are quite different as illustrated in Figure 4B. They also seem to be somewhat disconnected from the rest of the complex (more so for Cj), even though that's not apparent in Figures 2-4. Despite this, it would be very useful to show the

cryoEM densities when describing the catalytic site and C-terminal domain interactions, for example, as this can be very useful to increase confidence in the model and proposed mechanisms.

The description of the complex as a butterfly is engaging, and from a certain angle it can be made to look as such; this was also described previously in (Pillon et al., 2019, NSMB) for the same complex from a different organism (Ct). However, it is a bit misleading, because the complex is actually C2 symmetric. Under this symmetry, the 'body' would consist of two 'heads' one pointing up, one down facing towards the back, and one wing would have its back toward the viewer, the other the front. The structures presented here in Sc and Cj seem quite similar to the previous structure of the same complex in Ct, though the latter was only solved with cryoEM, and was also lacking the structure of the CC domain in Las1.

For the model suggested in Figure 8, perhaps in the 'weak activity' state, the LCT in Las1 could still be connected to Grc3, via the LCT, rather than disconnected as shown. This could facilitate faster assembly of the 'high activity' state. The complex is described as 'compact and stable', but from the structure and this image, it appears more dynamic, which would serve its purpose and the illustrated model better. The two copies of HEPN appear to have more connective area, meaning they are indeed more likely to remain assembled in the 'weak activity' state. On the other hand, HEPN in one protein appears to have less binding surface with PNK in Grc3, and even less so with the CTD (both PNK and CTD being from the other associated protein), meaning these bindings could release easily to form the 'weak activity' state.

There is also the potential to speculate that the GCT is bound to HEPN near the catalytic area in the 'weak activity' state. The reduced activity when the GCT residues are replaced by Alanine could then be explained by the complex not being able to assemble as quickly upon binding of the substrate, as it could if the GCT remained bound, rather than by a conformational change that it induces upon binding. The conformational change is also likely to be influenced by the combined binding of PNK and CTD in the assembled state, which also contact HEPN, rather than by GCT alone.

When comparing the structure of the HEPN domain in the lone Las1 protein to the structure of Las1-HEPN in the Las1-Grc3 complex, it is mentioned that 'large conformational changes are observed'. These could be described a bit better. The conformational change is ~3-4Å C-alpha RMSD across all ~150 residues in the domain (~90 residues forming a stable core that only changes by ~1Å). There is also a shift in the associated HEPN domain in Las1B domain compared to the bound HEPN in the Las1-Grc3 complex, as shown in Figure 7D: ~1Å shift and ~12degrees rotation. This does point to the conformation of HEPN changing upon complex formation, as does the relative positions of the HEPN domains in Las1A and Las1B. The conformational change and relative shift could indeed be key for the catalysis of the substrate as mentioned.

Overall, the structures presented should be very useful in further study of this system, even though the exact dynamics and how the substrate is bound are aspects that are perhaps not fully clear yet. The addition of the structures of the CC domain in two different organisms and the Las1 HEPN domain (not in complex with Grc3) as new structural information should allow for increasing our understanding of the overall complex and its mechanism.

Reviewer #2 (Public Review):

In this manuscript, Chen et al. determined the structural basis for pre-RNA processing by Las1-Grc3 endoribonuclease and polynucleotide kinase complexes from *S. cerevisiae* (Sc) and *C. jadinii* (Cj). Using a robust set of biochemical assays, the authors identify that the sc- and CjLas1-Grc3 complexes can cleave the ITS2 sequence in two specific locations, including a novel C2' location. The authors then determined X-ray crystallography and cryo-EM

structures of the ScLas1-Grc3 and CjLas1-Grc3 complexes, providing structural insight that is complimentary to previously reported Las1-Grc3 structures from *C. thermophilum* (Pillon et al., 2019, NSMB). The authors further explore the importance of multiple Las1 and Grc3 domains and interaction interfaces for RNA binding, RNA cleavage activity, and Las1-Grc3 complex formation. Finally, evidence is presented that suggests Las1 undergoes a conformational change upon Grc3 binding that stabilizes the Las1 HEPN active site, providing a possible rationale for the stimulation of Las1 cleavage by Grc3.

Several of the conclusions in this manuscript are supported by the data provided, particularly the identification and validation of the second cleavage site in the ITS2. However, several aspects of the structural analysis and complimentary biochemical assays would need to be addressed to fully support the conclusions drawn by the authors.

- There is a lack of clarity regarding the number of replicates performed for the biochemical experiments throughout the manuscript. This information is critical for establishing the rigor of these biochemical experiments.
- The authors conclude that Rat1-Rai1 can degrade the phosphorylated P1 and P2 products of ITS2 (lines 160-162, Figure 1H). However, the data in Fig. 1H shows complete degradation of 5'Phos-P2 and 5'Phos-P4 of ITS2, while the P1 and 5'Phos-P3 fragments remain in-tact. Additional clarification for this discrepancy should be provided.
- The authors determined X-ray crystal structures of the ScLas1-Grc3 (PDB:7Y18) and CjLas1-Grc3 (PDB:7Y17) complexes, which represents the bulk of the manuscript. However, there are major concerns with the structural models for ScLas1-Grc3 (PDB:7Y18) and CjLas1-Grc3 (PDB:7Y17). These structures have extremely high clashscores (>100) as well as a significant number of RSRZ outliers, sidechain rotamer outliers, bond angle outliers, and bond length outliers. Moreover, both structures have extensive regions that have been modeled without corresponding electron density, and other regions where the model clearly does not fit the experimental density. These concerns make it difficult to determine whether the structural data fully support several of the conclusions in the manuscript. A more careful and thorough reevaluation of the models is important for providing confidence in these structural conclusions.
- The presentation of the cryo-EM datasets is underdeveloped in the results section drawing and the contribution of these structures towards supporting the main conclusions of the manuscript are unclear. An in-depth comparison of the structures generated from X-ray crystallography and cryo-EM would have greatly strengthened the structural conclusions made for the ScLas1-Grc3 and CjLas1-Grc3 complexes.
- The authors conclude that truncation of the CC-domain contributes to Las1 IRS2 binding and cleavage (lines 220-222, Fig. 4C). However, these assays show that internal deletion of the CC-domain alone has minimal effect on cleavage (Fig 4C, sample 3). The loss in ITS2 cleavage activity is only seen when truncating the LCT and LCT+CC-domain (Fig 4C, sample 2 and 4, respectively). Consistently, the authors later show that Las1 is unable to interact with Grc3 when the LCT domain is deleted (Fig. 6 and Fig. 6-figure supplement 2). These data indicate the LCT plays a critical role in Las1-Grc3 complex formation and subsequent Las1 cleavage activity. However, it is unclear how this data supports the stated conclusion that the CC-domain is important for Las1 cleavage.
- The authors conclude that the HEPN domains undergo a conformational change upon Grc3 binding, which is important for stabilization of the Las1 active site and Grc3-mediated activation of Las1. This conclusion is based on structural comparison of the HEPN domains from the CjLas1-Grc3 complex (PDB:7Y17) and the structure of the isolated HEPN domain dimer (PDB:7Y16). However, it is also possible that the conformational changes observed in

the HEPN domain are due to truncation of the Las1 CC and CGT domains. A rationale for excluding this possibility would have strengthened this section of the manuscript.